TS5000 Robot Controller

Parameter Manual

Rev. 14

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IMPORTANT

- Make sure that this instruction manual is delivered to the final user of Shibaura Machine's industrial robot.
- Be sure to read this manual and that you fully understand its contents before operating the industrial robot.
- After reading this manual, keep it in an easily accessible place for future reference.

Shibaura Machine

This instruction manual is intended for the TS5000 controller.

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Introduction

Thank you for purchasing this SHIBAURA MACHINE robot system.
This manual describes the parameters for the TS5000 series robot controller and how to set them.
Various functions of the robot can be easily set and customized by setting parameters.

This manual consists of the following sections.

Chapter 1: Overview
This chapter gives an overview of parameter.

Chapter 2: Operation
This chapter describes how to display and operate the parameter operation screen.

Chapter 3: Setting
This chapter describes parameters.

Manufacturer

Shibaura Machine Co., Ltd.
2068-3, Ooka, Numazu-shi, Shizuoka-ken 410-8510, Japan
Tel: +81-55-926-5032
Fax: +81-55-925-6527

If you have purchased this controller outside of Japan, please contact the company that you purchased it from.

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Scope of Use

The product has been developed for use in a wide range of automation tasks including transport work such as the movement and alignment of workpieces and assembly work such as insertion of parts and screw tightening.
The customer should also use this product for the purpose of automating these tasks and for incorporating into automation equipment.

This product is not intended for use in tasks involving human contact (such as tasks for medical care and nursing care) or to be incorporated in equipment that sustains human life (such as life support devices and incubators), equipment on which people ride (including transportation equipment such as cars and aerospace equipment such as airplanes, and recreational equipment such as roller coasters), and equipment that protects human life or the human body (such as safety devices).
This product is not intended to be incorporated in equipment for controlling large heat quantities (such as nuclear power controllers and incinerator controllers).
This product is prohibited from being used for tasks or incorporated in equipment where a failure or malfunction could likely pose a direct threat to human life or harm to the human body. The customer is responsible for providing any warranties if this product is used in this way.

Usage Procedures

This section describes the operating conditions for TS5000. Usage under different conditions could expose the user to unexpected risks. Be sure to always use the controller in an environment that meets the required conditions.

1. Main specifications of the product: Refer to the robot manual for your robot.
2. Usage restrictions The “Restricted Specifications Sheet,” which shows the usage restrictions, are contained in Tables 1 to 6.

Table 1. Restricted Specifications Sheet

1. Purpose and application for using product (usage restrictions)	
1) Intended use	To make the robot perform operations for specific purposes (item transport, assembly, inspection, machining, etc.) (1) Transportation [1] Clear a transportation route to the unpacking site. [2] Use a forklift or other equipment to lift the packing box. [3] Move the packing box to the unpacking site. (2) Unpacking [1] Clear the space required for unpacking. [2] Cut the packing straps and unpack. [3] Cut the packing straps securing the robot. [4] Peel off the tape from the controller packing box, and unpack. [5] Check that the robot and controller are not damaged. [6] Check the accessories and their quantities. - When transferring the robot to the installation site [7] Dispose of, or store and manage the packing materials. - When storing the robot [7] Secure the robot in place in the same way as the state at delivery, and repack. [8] Move the packing box to the storage site. (3) Transfer - Robot [1] Check the orientation of the robot during transfer. [2] Check that the robot fixing jigs are mounted. [3] If using a crane, mount the supplied eyebolts. [4] Check the tightness of the fixing bolts of the robot fixing jig. [5] Use a crane or cart to transfer the robot to the installation site. - Controller [1] Use a cart to transfer the controller to the installation site.

Table 1. Restricted Specifications Sheet (continued)

1. Purpose and application for using product (usage restrictions)	
1) Intended use (continued)	(4) Installation [1] Use bolts to secure the robot to the frame and other parts. [2] Fabricate any needed cables. [3] Remove the robot fixing jigs and eyebolts. [4] Install the controller and connect it to the robot. [5] Press the robot by hand to move it to a location where it is easy to perform work. [6] Mount the end effector. [7] Connect the required external devices. (5) Teaching [1] Move the robot to the desired position or orientation by guiding it with a teach pendant or pushing it by hand. [2] Teach the position where the robot will operate. [3] Create the operation sequence (program, etc.). [4] Make sure that the signals and data are transferred to and from the external equipment. [5] Check operation at low speed (250 mm/sec or less). [6] While gradually increasing the operating speed, make sure that the operation matches the specific purpose. (6) Operation [1] Start robot operation. [2] Make sure that operation is being performed to match the specific purpose at the specified speed. [3] Stop robot operation. (7) Repacking [6] Remove the end effector. [2] Move the robot to the position and orientation where the fixing jig can be mounted by guiding it with a teach pendant or pushing it by hand. [3] Mount the fixing jig. [4] Turn off the power, and remove the wiring. [5] Obtain the space needed for packing. [6] If using a crane to transfer the robot, mount the supplied eyebolts. [7] Transfer the robot and controller to the packing site. [8] Pack the robot and controller so that they are in the original delivery state. [9] Clear a transportation route to the transportation equipment (truck, etc.). [10] Use a forklift to lift the packing box. [11] Load the packing box onto the transportation equipment.

Table 1. Restricted Specifications Sheet (continued)

1. Purpose and application for using product (usage restrictions)	
1) Intended use (continued)	(8) Maintenance Work performed when the primary power is off: - Robot [1] Check the bolts for any looseness and backlash and re-tighten them. [2] Make sure that the cable clamp is not loose. [3] Check for any damage. [4] Clean off any dust. [5] Inspect and apply corrosion inhibitor. [6] Check for any wear on the cable and air tube. [7] Check that there is no misalignment between the motion of each joint and the origin position. - Controller [1] Replace the memory backup batteries in the controller. Work performed when the primary power is on: [1] Check the greasing state and apply grease if needed (when the servo power is off). [2] Check for any abnormal vibrations and sounds during operation (when the servo power is on). [3] Check the teaching point for any misalignment (when the servo power is on). [4] Replace the backup batteries for the robot position detector (when the servo power is off). (9) Disposal [1] Remove the end effector. [2] Move the robot to the position and orientation where the fixing jig can be mounted by guiding it with a teach pendant or pushing it by hand. [3] Mount the fixing jig. [4] Turn off the power, and remove the wiring. [5] Use a crane to transfer the robot, and mount the supplied eyebolts. [6] Transfer the robot and controller to the disposal site. [7] Dispose of the robot and controller using a method compliant with your country’s laws and regulations.

Table 1. Restricted Specifications Sheet (continued)

1. Purpose and application for using product (usage restrictions)	
2) Foreseeable misuse	- Use for work which applies excessive impact to the robot or where excessive impact is applied by the robot - Modification of robot, controller, and optional parts, or supplying power while these covers are removed - Operation in other than the specified installation state (floor, suspended, and wall-mounted) - Operation on a frame or other structure that does not satisfy the specified strength - Operation under conditions not satisfying the specified installation requirements - Entry into the hazard zone of the robot by mistakenly assuming that a stop to wait for an operation signal during operation was an operation stop - Usage under conditions where the specification values are exceeded or usage under improper settings - Stop of the robot caused by power off during operation or when the servo power is on - Unintended operation caused by pressing a wrong button - Operating without checking operation in MANUAL mode - Operating without checking the tightness of the robot and hand set bolts - Unintended operation due to improper wiring - Unintended operation due to incorrect program selection - Riding on or leaning against the robot or hanging on the arm
3) Unexpected start	- Malfunction due to excessive noise - Robot start due to start signal unexpectedly sent from peripheral equipment to the robot - Unintended start due to abnormal communication data - Malfunction due to voltage fluctuation - Power on due to powering after power outage - Malfunction due to failure or the like

Table 2. Restricted Specifications Sheet

2. Replacement or other modification of program components (time restrictions)	
1) Mechanical restrictions	- Before operating the robot, perform operation at a low speed (20% or less of maximum speed) for five (5) minutes - Replace consumable parts as required or periodically by conducting daily inspection, regular inspection (every 3 and 6 months and every year) and overhauls (every 5 years).
2) Electrical restrictions	

Table 3. Restricted Specifications Sheet

3 Movable range of product (restriction on space)		
1) Work envelope	1st axis 2nd axis 3rd axis 4th axis Additional axis — —	Compliant with 1.4 Specifications Table Compliant with 1.4 Specifications Table Compliant with 1.4 Specifications Table Compliant with 1.4 Specifications Table Compliant with individual specifications. — —
2) Interface	Transportation: Unpacking: Transfer: Installation: Teaching: Operation: Repacking: Maintenance: Disposal:	The packing box shall be able to be transferred by a transportation worker. The packing box shall be able to be carried in. Space of at least 2 m^3 shall be provided. The robot shall be able to be hoisted by a crane. The robot shall be able to be lifted by two or more people and loaded onto a cart or similar device. There shall be no obstacles on the route from the unpacking site to the installation site. The product shall be installed at a location suitable for the outline and movable range of the robot. The product shall be installed at a location suitable for the outline and that provides adequate ventilation of the controller. The cables shall be installed at a location where they will not be stepped on by operators. Teaching work shall be able to be performed from outside the hazard zone of the robot. Space shall be provided so that, if a person enters the hazard zone, the person will not get entrapped by the robot or peripheral equipment. A fence or enclosure shall be installed to prevent any operator from approaching the hazard zone of the robot. The robot shall be able to be operated from outside the hazard zone of the robot. An operator in the fence or enclosure shall be visible from the operation location. Work shall be able to be performed to remove the robot hand, mount the fixing jig, separate the robot and controller, and remove the robot and controller from the frame. The robot shall be able to be moved from the installation site to the packing site. The repacking work shall be able to be performed. A forklift or other equipment shall be able to be used to lift the packing box. Space required for maintenance shall be provided. After maintenance, operation shall be able to be checked from outside the hazard zone of the robot. Space shall be provided so that, in the hazard zone, a person will not get entrapped by the robot or peripheral equipment. Work shall be able to be performed to remove the robot hand, mount the fixing jig, separate the robot and controller, and remove the robot and controller from the frame. The robot shall be able to be moved from the installation site to the disposal location.

Table 3. Restricted Specifications Sheet (continued)

3. Movable range of product (restrictions on space)	
3) Work environment	- A fence, enclosure, or other protection shall be provided that surrounds the hazard zone of the robot to prevent people from entering the area during operation. - A display indicating the robot operation status shall be provided to indicate whether the robot is operating or stopped. - Lighting shall be provided for enabling visual checks by operators. - To prevent operators from slipping, the floor shall not be covered with oil or similar substances. - No noises that impair attention shall be generated.

Table 4. Restricted Specifications Sheet

4. Life cycle of product
In the product life cycle of assembly, shipment testing, transportation, unpacking, transfer, installation, teaching, operation, repacking, maintenance, and disposal, the risk assessment is carried out for the life cycle of “transfer and installation” (transportation, unpacking, transfer, installation, and repacking), “use” (teaching, operation, and maintenance), and “disposal”.

Table 5. Restricted Specifications Sheet

5. Person who approaches the machine		
Applicable person	Knowledge and requirements	
1) Transportation worker	Knowledge	Contents of the instruction manual
	Requirement	Person who has received training for forklifts and other transfer equipment work as defined by the respective country's laws and regulations and who has been permitted to perform the work by the employer
2) Unpacking worker	Knowledge	Contents of the instruction manual
	Requirement	—
3) Transfer worker	Knowledge	Contents of the instruction manual
	Requirement	Person who has received training for crane work as defined by the respective country's laws and regulations and who has been permitted to perform the work by the employer
4) Installation worker	Knowledge	Contents of the instruction manual, operating procedures for target robot
	Requirement	When guiding a robot, a person who has received training for robot teaching work as defined by the respective country's laws and regulations and who has been permitted to perform the work by the employer
5) Teaching worker Work supervisor	Knowledge	Contents of the instruction manual, operating procedures for target robot, work regulations, and robot operations
	Requirement	Person who has received training for robot teaching work as defined by the respective country's laws and regulations and who has been permitted to perform the work by the employer
6) Operator	Knowledge	Contents of the instruction manual, operating procedures for target robot, work regulations, and robot operations
	Requirement	—
7) Repacking worker	Knowledge	Contents of the instruction manual, operating procedures for target robot
	Requirement	Person who has received training for crane work as defined by the respective country's laws and regulations and who has been permitted to perform the work by the employer When guiding a robot, a person who has received training for robot teaching work as defined by the respective country's laws and regulations and who has been permitted to perform the work by the employer

Table 5. Restricted Specifications Sheet (continued)

5. Person who approaches the machine		
Applicable person	Knowledge and requirements	
8) Maintenance worker Work supervisor	Knowledge	Contents of the instruction manual, operating procedures for target robot, work regulations, and robot operations Storage procedures for target robot
	Requirement	Person who has received training for robot inspection work as defined by the respective country's laws and regulations and who has been permitted to perform the work by the employer
9) Disposal worker	Knowledge	Contents of instruction manual, operating procedures for target robot, and waste disposal methods
	Requirement	Person who has received training for crane work as defined by the respective country's laws and regulations and who has been permitted to perform the work by the employer When guiding a robot, a person who has received training for robot teaching work as defined by the respective country's laws and regulations and who has been permitted to perform the work by the employer
10) People passing nearby Other workers	- Measures shall be taken to prevent entry into the fence, enclosure, or other protection surrounding the hazard zone of the robot. - Measures shall be taken to make the robot stop operation when someone approaches the hazard zone. - If a person enters the hazard zone of the robot to perform work, robot operation shall be stopped, the drive power turned off, and a teaching worker or maintenance worker shall supervise the situation to ensure safety.	

Table 6. Restricted Specifications Sheet

6. Other

Warranty

We conduct strict testing and inspection of this product before delivery to ensure that it satisfies our performance standards.
We warrant this product to be free from defects under the following terms and conditions.

1. Warranty period

The warranty period of the product is either of the following, whichever is earlier.

(1) 18 months after shipping from our factory
(2) 12 months after installation at customer site

2. Warranty scope

(1) The warranty is provided for this product. The warranty covers only the specifications and functions included in the product specifications, catalog, and instruction manual. Any secondary or incidental damages caused by failure of this product are not covered by the warranty under any circumstances.
(2) We will repair a faulty product free of charge only if this product has been operated and used in accordance with the instruction manual supplied with the product.
A fee will be charged for any repairs after the warranty period has expired.

3. Disclaimer

The warranty shall not apply to the following cases.

(1) Any failure or damage arising from careless use or misuse, or usage that is not described in the instruction manual.
(2) Any failure caused by aging or normal wear and tear (natural fading of paints, deterioration of consumable parts[1)].
(3) Irregularities confined to sensory quality such as noise not recognized as affecting functions
(4) When any modification or disassembly without our consent was performed
(5) Any failure or damage arising from inadequate maintenance and inspection or improper repair
(6) Any failure or damage due to an act of God, fire, and other force majeure
(7) Internal data such as programs and points created or modified by customers
(8) When a product purchased in Japan is taken abroad

*1): Consumable parts refer to the replacement parts for maintenance as listed in the manual of the controller.

4. Safety Information

(1) Shibaura Machine shall provide no guarantees for the standard performance of this product when used outside the specifications.
(2) If the customer does not observe the safety information appearing for DANGER, WARNING, CAUTION, and NOTES contained in this instruction manual, Shibaura Machine will not be liable for any resulting injuries, death, or other accidents, damage, product failure, or any other problems.
(3) Note that the safety information appearing for DANGER, WARNING, CAUTION, and other explanations in this instruction manual are within the scope of usage assumed by Shibaura Machine, and they do not foresee all possible incidents for all situations that could occur.

Standards, laws, and regulations pertaining to the safety of industrial robots

The international industrial standard ISO/DIS 12100 "Safety of machinery" specifies risk reduction measures for machinery and equipment.
Step 1: Inherently safe design measures - Limitations on force, speed, energy, etc.
Step 2: Safeguarding - Installation of safety fences, etc.
Complementary protective measures - Installation of emergency stop devices, etc.
Step 3: Information for use - Warning signs, alarms, instruction manuals, etc.
Based on the above, the ISO/IEC has defined "Guide 51", which classifies the various standards in a hierarchal structure.

Safety standards for industrial robots are standardized as separate safety standards for machinery in Type C.

- ISO 10218, ISO 10218-1

Be sure to always follow the standards, laws, and regulations set forth in the country where the industrial robot is used.

Safety Information

This instruction manual contains important information to ensure the safe and correct use of the controller and to prevent injury to the operator and others and damage to property.
Be sure that you fully understand the following information (signal words and symbols) before reading the main text, and always follow the provided safety instructions.

Display of Signal Words and Their Meanings

Signal Word and Symbol	Meaning
⚠DANGER	Indicates an imminently hazardous situation which, if not avoided, will result in death or serious injury.
⚠WARNING	Indicates a potentially hazardous situation which, if not avoided, will result in death or serious injury.
⚠CAUTION	Indicates the possibility that incorrect operation could cause injury[1)] to the operator.
NOTICE	Indicates the possibility that incorrect operation could cause damage to property[2)] and provides helpful advice about performing work.

1) Injury refer to wounds, burns, electric shocks, and similar cases that do not require hospitalization or prolonged outpatient treatment.
2) Damage to property refers to extensive damage to property or materials.

⚠DANGER
●Be sure to carefully read the safety manual described in the target controller manual in order to ensure the safety of the controller from transportation to disposal. Performing an operation different from what is described in the manual could result in harm to people.

Service Network

Please refer to our distributor's handouts

Overview

This chapter describes the terms used in this manual.

1. Parameter Overview

Overview

Parameters are changed to set and customize various functions of the robot easily.
The parameters are described in the parameter file, and parameter setting are performed by changing them.
The parameter configuration and description method are explained in the following.

Configuration

Parameters are configured as follows.

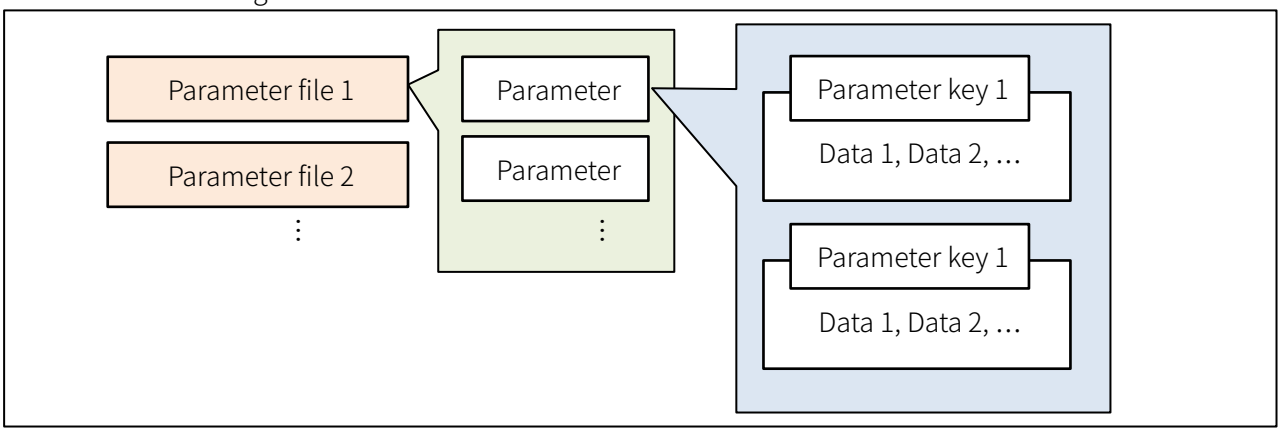


Figure 1.1 Parameter file configuration

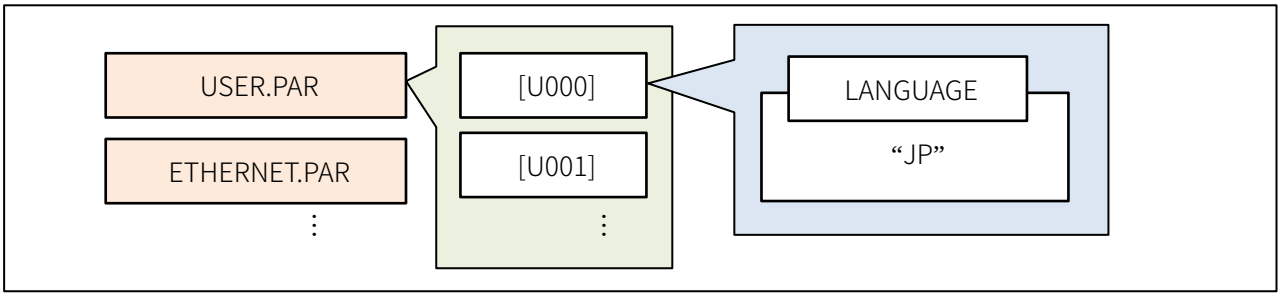


Figure 1.2 Example of parameter file configuration

⚠ WARNING

- Do not delete the parameter file. Doing so may result in a collision with the robot that has performed an unexpected motion. If the parameter file is deleted, the robot makes motion in the setting different from the one before deletion.
- Do not change the parameter file name. Doing so may result in a collision with the robot that has performed an unexpected motion. If the parameter file name is changed, the robot makes motion in the setting different from the one before change.

NOTICE

- Never set the values other than the ones described in the parameter manual. Doing so may result in a collision between the robot that has performed an unexpected motion and a peripheral device.

Parameter

Parameters are described as follows.

```
‘[U000] LANGUAGE
LANGUAGE = “JP”
‘[U001] SYSTEM BASE COORDINATE
SYS-BASE-COOD= 500.000000, 100.000000, 50.000000, 0.000000
```

A number and name are assigned to each parameter.

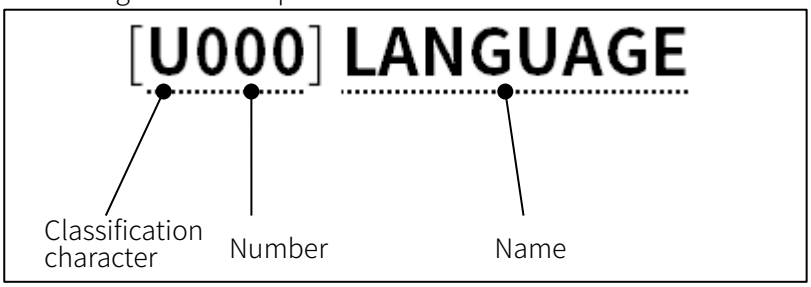


Figure 1.3 Example of parameter number and name

A classification character precedes the beginning of the number, and is assigned to each parameter file as follows.

Parameter file	Classification character
USER.PAR	U
ETHERNET.PAR	E
PLC.PAR	P
EXTRNSEL.PAR	X
VISION.PAR	V
AREA.PAR	A

A parameter has one or more parameter keys.
Also, a parameter key has parameter data.
Changing a parameter refers to changing parameter data that a parameter key has.

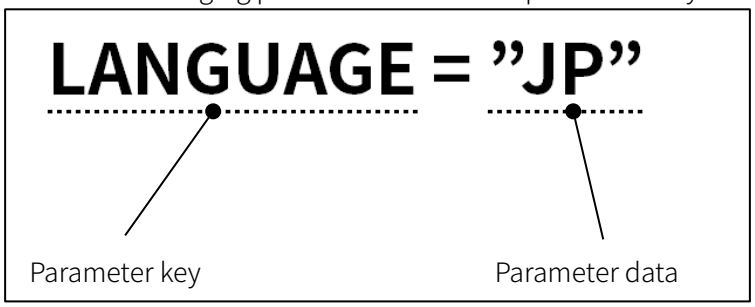


Figure 1.4 Example of parameter key and parameter data

Special description

If there are multiple parameter data items for a parameter key, the parameter key and parameter data can also be described as follows.

Example

```
SYS-BASE-COOD = 500.000000, 100.000000, 50.000000, 0.000000
```

↓

```
SYS-BASE-COOD[0] = 500.0000
SYS-BASE-COOD[1] = 100.0000
SYS-BASE-COOD[2] = 50.00000
SYS-BASE-COOD[3] = 0.000000
```

Comment

As with the SCOL language, a single quotation (‘) can be used to add a comment to the parameter file. The comment can be used to edit the parameter file directly using TP1000, etc.

Example

```
‘comment
PLUS-SOFT-LIMIT = 116.0, 141.0, 121.0, 361.0, 0.000
```

MEMO

The comment following parameter data disappears due to parameter setting operation using TP5000 or TSAssist.

To avoid this,

instead of specifying 116.0, 141.0, 121.0, 361.0, 0.000 ‘comment,

describe a comment on a different line as follows.

‘comment

PLUS-SOFT-LIMIT = 116.0, 141.0, 121.0, 361.0, 0.000

Operation

This chapter describes the operation method for perform parameter setting.

2. Opening the Parameter Setting Screen

This section explains how to open the parameter setting screen.

The parameter setting screen is used to change a parameter easily.

Parameter setting can be performed without need to consider any parameter key.

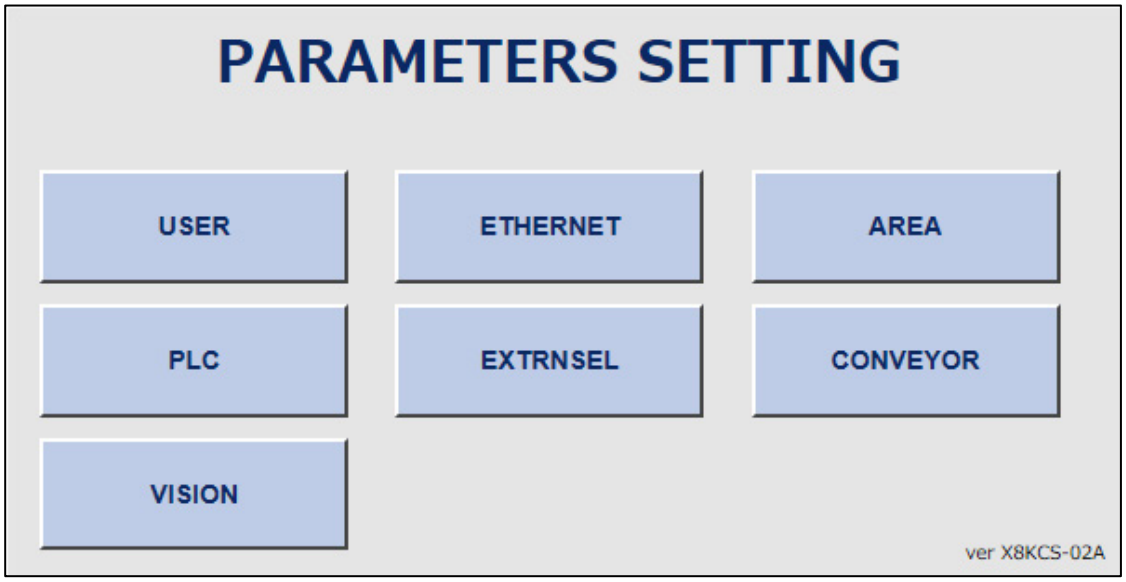


Figure 2.1 Parameter setting screen

There are two ways to open the parameter setting screen:

(1) USING TP5000 to open the parameter setting screen

(2) Using TSAssist to open the parameter setting screen

Using TP5000 to open the parameter setting screen

This section describes the operation method for using TP5000 to perform parameter setting.

1 Display the top screen.

> *MEMO*
>
> *For details about the method for displaying the top screen from the current screen, refer to STA1161_ TP5000 OPERATION MANUAL.*

2 Switch the master mode key for TP5000 to “MANUAL.”

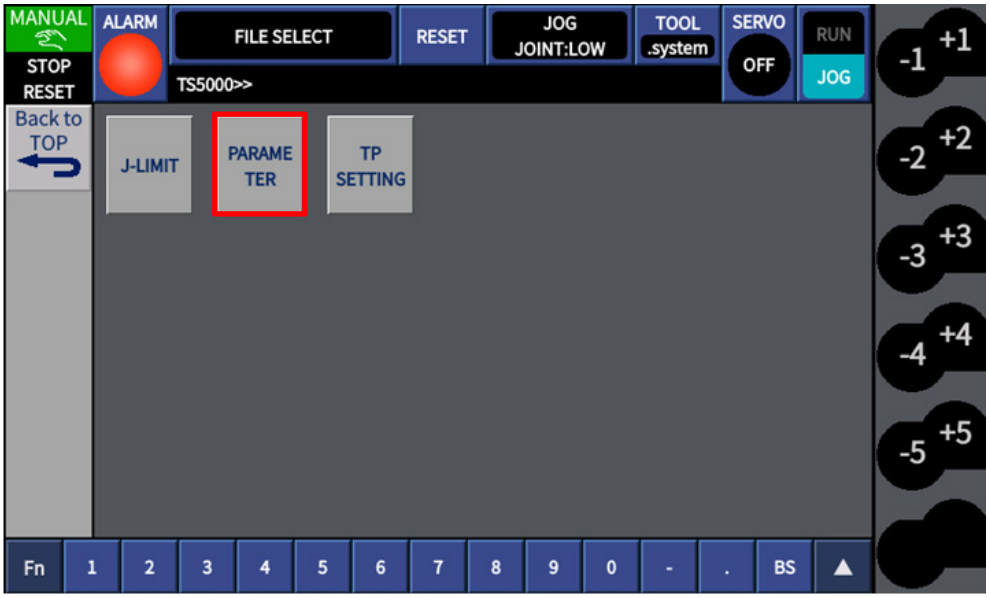
3 Tap the SETTING button.

The setting menu screen appears.



4 Tap the parameter button.

The parameter setting screen appears.



Using TSAssist to open the parameter setting screen

This section describes the operation method for using TSAssist to perform parameter setting.

1 Start TSAssist.

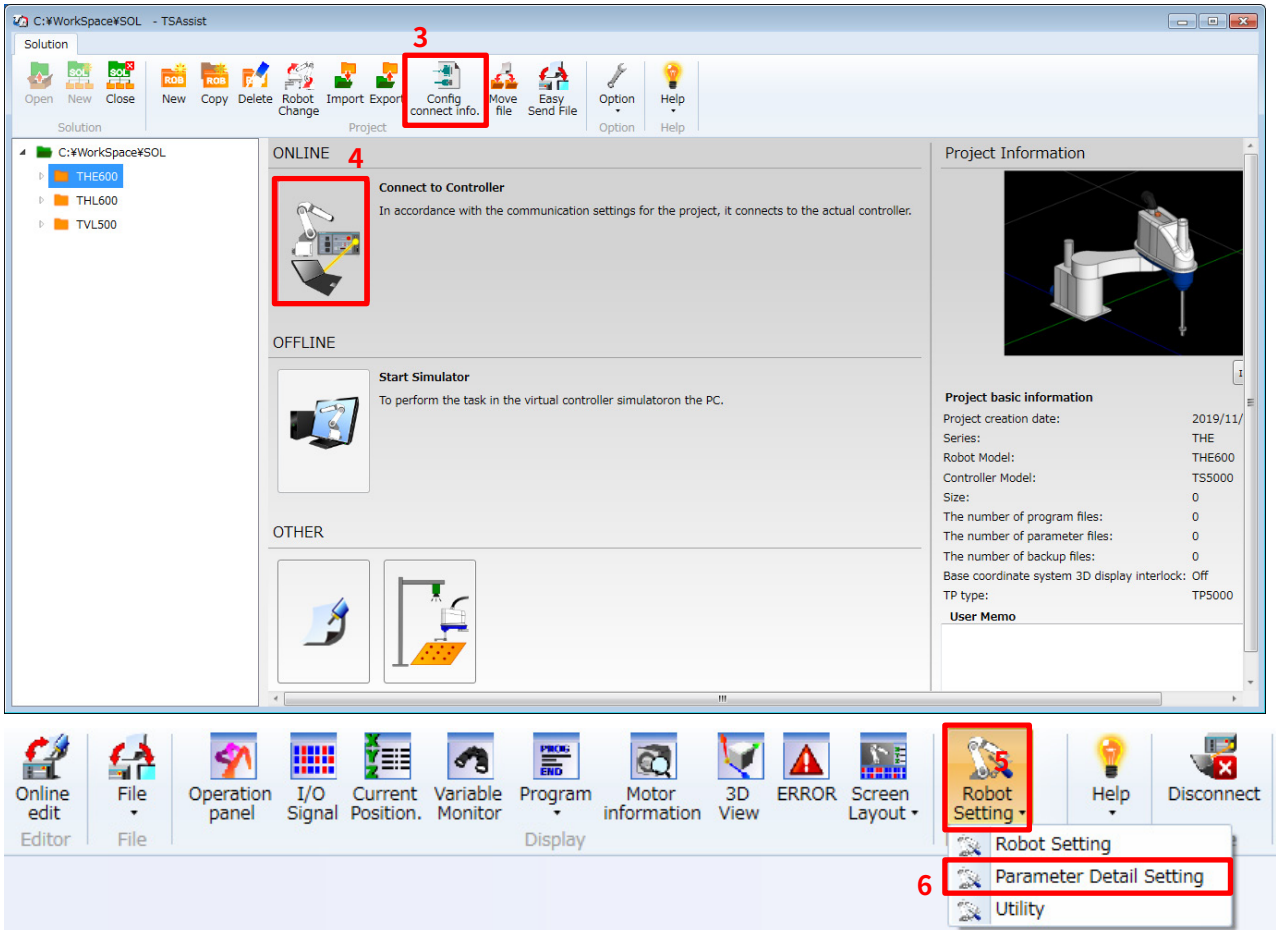
2 Connect TSAssist and the controller via the Ethernet cable.

3 Press [Config connect info.] to perform communication setting.

4 Select [Connect to Controller] and connect TSAssist and the controller online.

5 Click [Robot Setting] in the ribbon menu at the upper part of the screen.

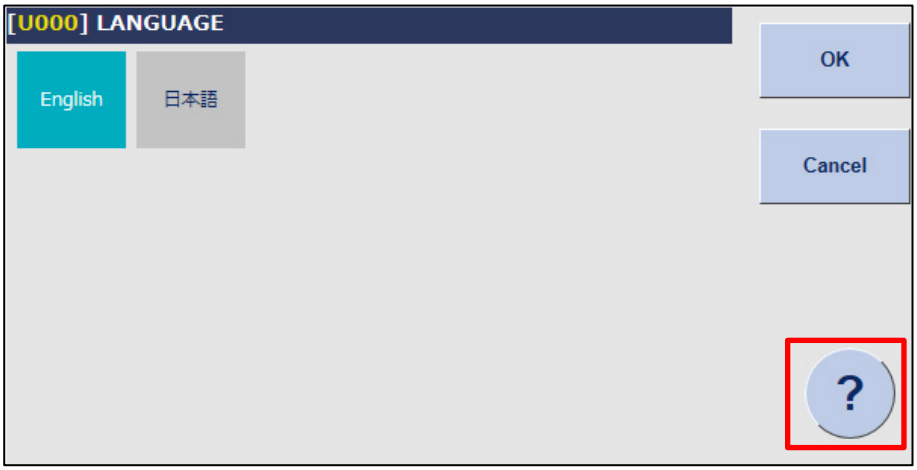
6 Click [Parameter Detail Setting] in the pull-down menu.



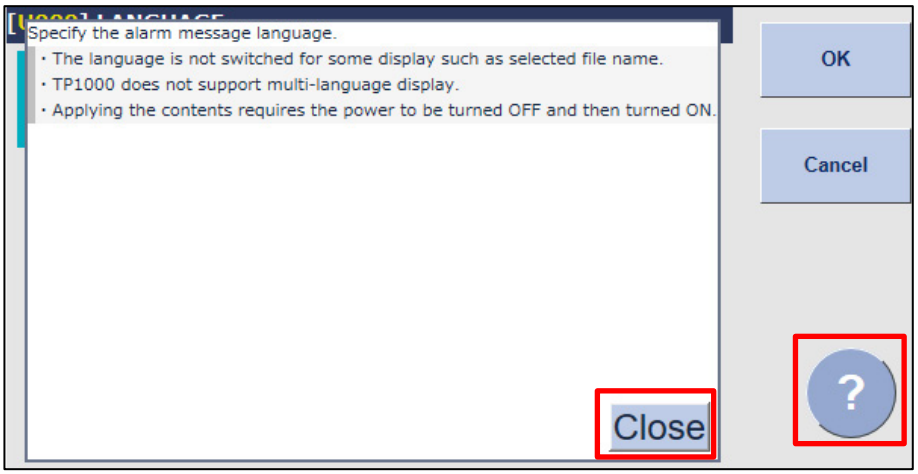
Displaying Help on the parameter setting screen

An explanation of the parameter can be displayed for each item on the parameter setting screen.

To display the explanation, click the [?] button at the bottom right of the parameter screen.



To close the parameter help, click the [Close] or [?] button.



MEMO

- *This function is supported by the controller version X8KCS-02A or higher.*
- *Help is displayed in English only.*
- *Help is not displayed on some screens.*

3. Using TP1000 to edit parameters

This section describes the operation method for using TP1000 to perform parameter setting.
When using TP1000, the parameter setting screen cannot be opened; therefore, parameter files need to be edited from the file edit screen.

1 Connect TP1000 with the controller.

MEMO

For details about how to connect TP1000 and the controller, refer to "STA1169_TP1000 OPERATION MANUAL."

2 Switch the master mode for the controller to MANUAL mode.

MEMO

For details about how to switch the master mode to MANUAL mode, refer to "STA1169_TP1000 OPERATION MANUAL."

3 Press the NEXT key until "EDIT" in the function menu on the basic screen.

```
MODE:MANUAL    /CON    TEST
                                  L:3
                                  O:20%
                                  M:FREE
                                  S:STOP
                                    RESET

 EDIT   DEDIT   SEL    RUN    FILE
```

4 Press "EDIT" (F1) key.

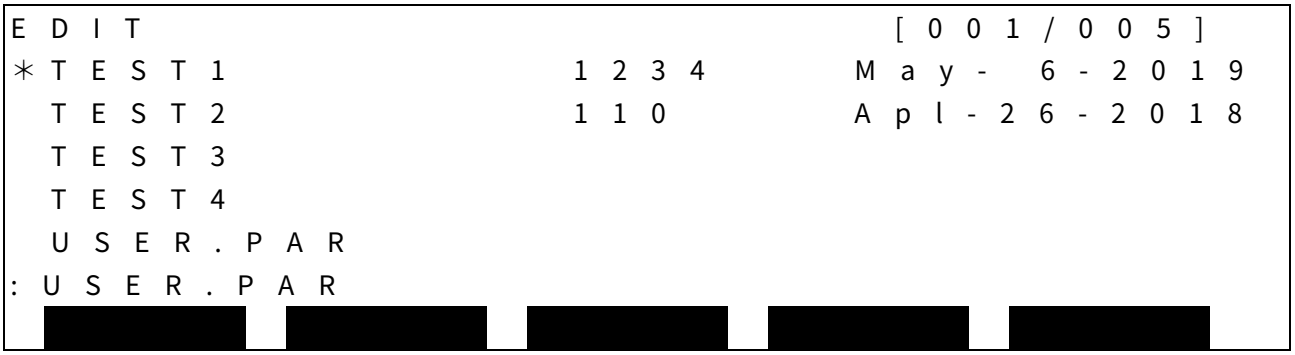
The file list screen appears.

```
EDIT                                [001/005]
*                    1234      May- 6-2019
  TEST2              110       Apl-26-2018
  TEST3
  TEST4
  USER.PAR

 NAME   TIME   NEWA
```

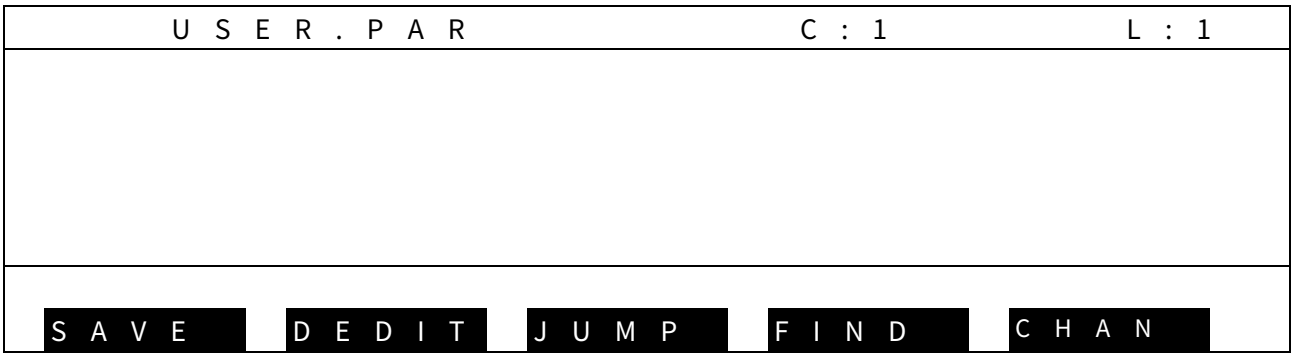
5 Select the parameter file to be changed from the file list, and then press the [EXE] key.

The parameter file name appears on the command line.



6 Press the [EXE] key.

File edit screen appears.



MEMO

If TP1000 has been used to perform parameter setting, applying the setting contents requires the power to be turned OFF and then turned ON.

Setting

This chapter describes the following parameters.

USER.PAR
ETHERNET.PAR
PLC.PAR
EXTRNSEL.PAR
VISION.PAR
AREA.PAR

USER.PAR

This section explains USER.PAR.

USER.PAR sets basic operation and functions.

[U000] LANGUAGE

Set language

Specify the alarm message language.

Parameter key	LANGUAGE	
Parameter data	=(language)	
Data type	String type	
Number of data items	1	
Data range	”JP” or ”EN” or ”CN”	
Description	=”JP”	Set the language to Japanese.
	=”EN”	Set the language to English.
	=”CN”	Set the language to Chinese.
Default	=”EN”	

Setting example

```
‘[U000] LANGUAGE
LANGUAGE = “JP”
```

MEMO

- *The language is not switched for some display such as selected file name.*
- *TP1000 does not support multi-language display.*
- *Applying the contents requires the power to be turned OFF and then turned ON.*

[U001] SYSTEM BASE COORDINATE

Set BASE coordinate system

Specify the base coordinate system when the controller is started.
If the robot has been relocated, correct the deviation in teach point.

Parameter key	SYS-BASE-COOD
Parameter data	=(Axis 1), (Axis 2), (Axis 3), (Axis 4)
Data type	Real number type
Number of data items	4
Data range	-9999.999 to 9999.999 (mm)
Default	=0.0, 0.0, 0.0, 0.0

Setting value

```
‘[U001] SYSTEM BASE COORDINATE
SYS-BASE-COOD = 010.000000, 0500.000000, 010.000000, 0.000000
```

MEMO

・The base coordinate can be changed by using the BASE command on a program. In this case, the parameter is not rewritten. If the program is reset, the base coordinate system returns to the value set in [U001].

・Applying the contents required the power to be turned OFF and then turned ON.

NOTICE

- If this parameter has been changed after teaching work, confirm the teach point.
  The robot that has made an unexpected motion may collide with a peripheral device.

[U002] SOFTWARE LIMIT

Set software limit

Specify the movable range of joint axes by software. When the set movable range of joint axes is exceeded, software limit alarms 4-42* and 4-43* occur.

Parameter key	PLUS-SOFT-LIMIT
Parameter data	=(Axis 1), (Axis 2), (Axis 3), (Axis 4), (Axis 5)
Data type	Real number type
Number of data items	5
Data range	-9999.999 to 9999.999 (deg)
Description	Set software limit in the plus direction of each joint.
Default	=9999.0, 9999.0, 9999.0, 9999.0, 9999.0, 9999.0, 9999.0, 9999.0

Parameter key	MINUS-SOFT-LIMIT
Parameter data	=(Axis 1), (Axis 2), (Axis 3), (Axis 4), (Axis 5)
Data type	Real number type
Number of data items	5
Data range	-9999.999 to 9999.999 (deg)
Description	Set software limit in the minus direction of joints.
Default	=-9999.0, -9999.0, -9999.0, -9999.0, -9999.0, -9999.0, -9999.0, -9999.0

Setting example

```
'[U002] SOFTWARE LIMIT
PLUS-SOFT-LIMIT = 116.0, 141.0, 121.0, 361.0, 0.000
MINUS-SOFT-LIMIT = -116.0, -141.0, -1.0, -361.0, 0.000
```

MEMO

- *The unit of SCARA robot axis 3 is (mm).*

NOTICE

- Be sure to set the software limit inside the mechanical stopper.
  If the software limit is not set inside the mechanical stopper, the robot in motion may collide with the mechanical stopper.
- After software limit is set, be sure to confirm that the software limit alarms (4-42* and 4-43*) occur correctly when the joint is moved beyond the range.

[U003] ACCEL SUPPRESS FUNCTION SW

Suppress SPEED command acceleration

When this function is disabled (= 0), acceleration is suppressed by changing the motion speed using the SPEED command.
Enabling this function (= 1) keeps acceleration constant without changing the motion speed using the SEEP command. This can shorten the movement time due to acceleration/deceleration of the robot.

Parameter key	ACCEL-SUPPRESS-SW	
Parameter data	=(disabled/enabled)	
Data type	Integer type	
Number of data	1	
Data range	0 or 1	
Description	=0	Acceleration is suppressed using the SPEED command.
	=1	Acceleration is not suppressed using the SPEED command.
Default	=0	

Setting example

```
'[U003] ACCEL SUPPRESS FUNCTION SW
ACCEL-SUPPRESS-SW    = 1
```

MEMO

For details about the change factor of acceleration and deceleration of the robot, refer to STA1163_ROBOT LANGUAGE MANUAL.

[U004] DEFAULT JOG SETTING

Initial value of manual guidance

Specify the manual guidance setting after the controller power is turned on.

Parameter key	DEF-JOG-SETTING
Parameter data	=(guidance coordinate), (jog speed), (inching size), (guidance mode)
Data type	Integer type
Number of data items	4
Data range	It varies depending on parameter data. See below.
Default	=0,0,0,0

(Guidance coordinate)

Specify the guidance coordinate selected as default.

Description	=0	Joint axis
	=1	Tool
	=2	Work
	=3	World

(Jog speed)

Specify the jog operation speed selected as default.

Description	=0	Low
	=1	Mid
	=2	High

(Inching size)

Specify the inching size selected as default.

Description	=0	Small
	=1	Medium
	=2	Large

(Guidance mode)

Specify the guidance mode selected as default.

Description	=0	JOG
	=1	INCHING
	=2	FREE

Setting example

```
‘[U004] DEFAULT JOG SETTING
DEF-JOG-SETTING= 0, 0, 0, 0
```

MEMO

Applying the contents requires the power to be turned OFF and then turned ON.

[U005] JOG SPEED CHANGE FUNCTION SW

Set manual guidance

Enable the settings of jog speed in [U006], inching speed in [U007], and inching amount in [U008].

Parameter key	JOG-SPEED-SW	
Parameter data	=(Disabled/Enabled)	
Data type	Integer type	
Number of data items	1	
Data range	0 or 1	
Description	=0	The robot operates according to the internal parameter[1] that was set at shipment.
	=1	Enable the settings of [U006], [U007], [U008].
Default	=0	

1) An internal parameter is a parameter set at factory shipment. The value of an internal parameter varies depending on the robot.

Setting example

```
‘[U005] JOG SPEED CHANGE FUNCTION
JOG-SPEED-SW = 1
```

MEMO

Teach point movement and BYPASS operate regardless of this parameter.

NOTICE

- If this parameter has been enabled, confirm the set values in[U006], [U007], and [U008].
  The robot that has made an unexpected motion may collide with a peripheral device.

[U006] JOG SPEED

Set jog speed

Specify the speed at which the robot is moved due to jog operation by percentage (%) using the maximum speed as a base of 100.
Using this function requires [U005] to be enabled.

Setting when guidance coordinate is world coordinate system, work coordinate system and tool coordinate system

Parameter key	JOG-HIGH-S
Parameter data	=(Movement speed)
Data type	Integer type
Number of data items	1
Data range	1 to internal parameter value (%)
Description	Specify the movement speed at high jog speed.
Default	=3

Parameter key	JOG-MID-S
Parameter data	=(Movement speed)
Data type	Integer type
Number of data items	1
Data range	1 to internal parameter value (%)
Description	Specify the movement speed at mid jog speed.
Default	=2

Parameter key	JOG-LOW-S
Parameter data	=(Movement speed)
Data type	Integer type
Number of data items	1
Data range	1 to internal parameter value (%)
Description	Specify the movement speed at low jog speed.
Default	=1

Setting when guidance coordinate is joint axis

Parameter key	JOG-HIGH-J
Parameter data	= (Axis 1), (Axis 2), (Axis 3), (Axis 4), (Axis 5), (Axis 6), (Axis 7), (Axis 8)
Data type	Integer type
Number of data items	8
Data range	1 to internal parameter value (%)
Description	Specify the movement speed at high jog speed.
Default	=3, 3, 3, 3, 3, 3, 3, 3

Parameter key	JOG-MID-J
Parameter data	= (Axis 1), (Axis 2), (Axis 3), (Axis 4), (Axis 5), (Axis 6), (Axis 7), (Axis 8)
Data type	Integer type
Number of data items	8
Data range	1 to internal parameter value (%)
Description	Specify the movement speed at mid jog speed.
Default	=2, 2, 2, 2, 2, 2, 2, 2

Parameter key	JOG-LOW-J
Parameter data	= (Axis 1), (Axis 2), (Axis 3), (Axis 4), (Axis 5), (Axis 6), (Axis 7), (Axis 8)
Data type	Integer type
Number of data items	8
Data range	1 to internal parameter value (%)
Description	Specify the movement speed at low jog speed.
Default	=1, 1, 1, 1, 1, 1, 1, 1

Setting example

```
‘[U006] JOG SPEED
JOG-HIGH-S = 3
JOG-MID-S = 5
JOG-LOW-S = 1
JOG-HIGH-J = 3, 3, 3, 3, 3, 0, 0, 0
JOG-MID-J = 2, 2, 2, 2, 2,0, 0, 0
JOG-LOW-J = 1, 1, 1, 1, 0, 0, 0, 0
```

MEMO

- *If the value is set to 0, the operation is performed at the speed specified in the internal parameter which was set at shipment.*
- *Set the movement speed so that it becomes low speed < mid speed < high speed.*

[U007] INCHING SPEED

Set inching speed

Specify the speed at which the robot is moved due to inching operation by percentage (%) using the maximum speed as a base of 100.
Specify the movement speed at which inching operation is performed in the world coordinate system, work coordinate system and tool coordinate system, and at which inching operation for joint axes is performed.
Using this function requires [U005] to be enabled.

Parameter key	INCH-SPD-S
Parameter data	= (Movement speed)
Data type	Integer type
Number of data items	1
Data range	1 to internal parameter value (%)
Description	Specify the movement speed at which inching operation is performed in the world coordinate system, work coordinate system and tool coordinate system.
Default	=10

Parameter key	INCH-SPD-J
Parameter data	= (Axis 1), (Axis 2), (Axis 3), (Axis 4), (Axis 5), (Axis 6), (Axis 7), (Axis 8)
Data type	Integer type
Number of data items	8
Data range	1 to internal parameter value (%)
Description	Specify the movement speed at which inching operation for joint axes is performed.
Default	=10, 10, 10, 10, 10, 10, 10, 10,

Setting example

```
‘[U009] INCHING SPEED
INCH-SPD-S = 10
INCH-SPD-J = 5, 5, 5, 5, 5, 0, 0, 0
```

MEMO

- *If the inching speed is set to 0, the operation is performed at the speed specified in the internal parameter which was set at shipment.*

[U008] INCHING AMOUNT

Set inching amount

Specify the amount of movement of the robot due to inching operation by (mm) or (deg).
Using this function requires [U005] to be enabled.
If a minute movement amount is specified, the robot may not operate according to the specified value.

Setting when guidance coordinate is world coordinate system, work coordinate system or tool coordinate system

Parameter key	INCH-LARG-S
Parameter data	= (Movement amount)
Data type	Real number type
Number of data items	1
Data range	0.001 to 10.00(mm)
Description	Specify the movement amount when inching amount is large.
Default	=1.0

Parameter key	INCH-MID-S
Parameter data	= (Movement amount)
Data type	Real number type
Number of data items	1
Data range	0.001 to 10.00(mm)
Description	Specify the movement amount when inching amount is mid.
Default	=0.5

Parameter key	INCH-SMAL-S
Parameter data	= (Movement amount)
Data type	Real number type
Number of data items	1
Data range	0.001 to 10.00(mm)
Description	Specify the movement amount when inching amount is small.
Default	=0.1

Setting when guidance coordinate is joint axis

Parameter key	INCH-LARG-J
Parameter data	= (Axis 1), (Axis 2), (Axis 3), (Axis 4), (Axis 5), (Axis 6), (Axis 7), (Axis 8)
Data type	Real number type
Number of data items	8
Data range	0.001 to 5.00(deg), 0.001 to 10.00 (mm) for Axis 3 only
Description	Specify the movement amount when the inching amount of joint axes is large.
Default	=0.10, 0.10, 0.10, 0.10, 0.10, 0.10, 0.10, 0.10

Parameter key	INCH-MID-J
Parameter data	= (Axis 1), (Axis 2), (Axis 3), (Axis 4), (Axis 5), (Axis 6), (Axis 7), (Axis 8)
Data type	Real number type
Number of data items	8
Data range	0.001 to 5.00(deg), 0.001 to 10.00 (mm) for Axis 3 only
Description	Specify the movement amount when the inching amount of joint axes is mid.
Default	=0.05, 0.05, 0.05, 0.05, 0.05, 0.05, 0.05, 0.05

Parameter key	INCH-SMAL-J
Parameter data	= (Axis 1), (Axis 2), (Axis 3), (Axis 4), (Axis 5), (Axis 6), (Axis 7), (Axis 8)
Data type	Real number type
Number of data items	8
Data range	0.001 to 5.00(deg), 0.001 to 10.00 (mm) for Axis 3 only
Description	Specify the movement amount when the inching amount of joint axes is small.
Default	=0.01, 0.01, 0.01, 0.01, 0.01, 0.01, 0.01, 0.01

Setting example

```
‘[U010] INCHING AMOUNT
INCH-LARG-S = 1.000000
INCH-MID-S = 0.500000
INCH-SMAL-S = 0.100000
INCH-LARG-J = 0.100000, 0.100000, 0.100000, 0.100000, 0.100000, 0.100000, 0.100000, 0.100000
INCH-MID-J = 0.050000, 0.050000, 0.050000, 0.050000, 0.050000, 0.050000, 0.050000, 0.050000
INCH-SMAL-J = 0.010000, 0.010000, 0.010000, 0.010000, 0.010000, 0.010000, 0.010000, 0.010000
```

MEMO

The unit of SCARA robot axis 3 is (mm).

If data is set to 0, the operation is performed with the inching amount that was set at shipment.

[U009] JOG PAYLOAD

Set PAYLOAD during jog operation

Specify the PAYLOAD value during jog operation.

Parameter key	MANUAL-PAYLD
Parameter data	= (Mass[Kg]), (Offset[mm])
Data type	Real number type
Number of data items	2
Data range	0 or more
Default	=0.0, 0.0

(Mass[kg])
Specify the weight of the load on the robot tip in units of kilograms (kg).

(Offset[mm])
Specify the distance of the gravity center of the load on the robot tip from the center of the tool mounted on the robot tip in units of millimeters (mm).

Setting example

```
‘[U009] JOG PAYLOAD
MANUAL-PAYLD = 0.000000, 0.000000
```

MEMO

To perform automatic operation, be sure to use the PAYLOAD command.

The PAYLOAD command format is

PAYLOAD= {M, L}

where the tool mass is M (kg) and gravity center offset is L (mm).

For details, refer to STA1163_ROBOT LANGUAGE MANUAL (SCOL2).

NOTICE

- If the robot vibrates during guidance operation, refer to the target robot manual and be sure to perform PAYLOAD setting during manual operation. Vibration deteriorates the performance and lifetime of the robot.
  - When the load mass and gravity center offset are large, the robot vibrates.

[U010] MOVES TO FUNCTION SW

Set movement method for teach point

When enabled (=1), this function changes the motion command in teach point movement. This function sees the guidance coordinate status to switch the motion command.
This function can be used on TP1000 only.

Parameter key	MOVES-TO-SW	
Parameter data	= (Disabled/Enabled)	
Data type	Integer type	
Number of data items	1	
Data range	0 or 1	
Description	=0	All MOVE operation
	=1	When the guidance coordinate is joint axis, operation is performed by the MOVE command. When the guidance coordinate is world, tool, and work, operation is performed by the MOVES command.
Default	=0	

Setting example

```
‘[U010] MOVES TO FUNCTION SW
MOVES-TO-SW = 1
```

MEMO

In BYPASS teach point movement, the motion command is also switched similarly.

NOTICE

- Be careful about change in the movement route of the teach point due to change of this parameter. The robot that has made an unexpected motion may collide with a peripheral device.

[U011] BYPASS FUNCTION

Set BYPASS operation

Perform BAYPASS (move to teach point bypass).
This function can be used on TP1000 only.
In the SCARA robot, when the robot moves to the teach point, it once retracts its hand to prevent collision with a workpiece present midway.

If this function is enabled, the robot moves from current position P0 to teach point P1 in the following order.

1. Moves by retracting from current position P0 to P0a.
2. Parallel move + rotation from position P0a to P1a.
3. Moves by extending from position P1a to P1a.

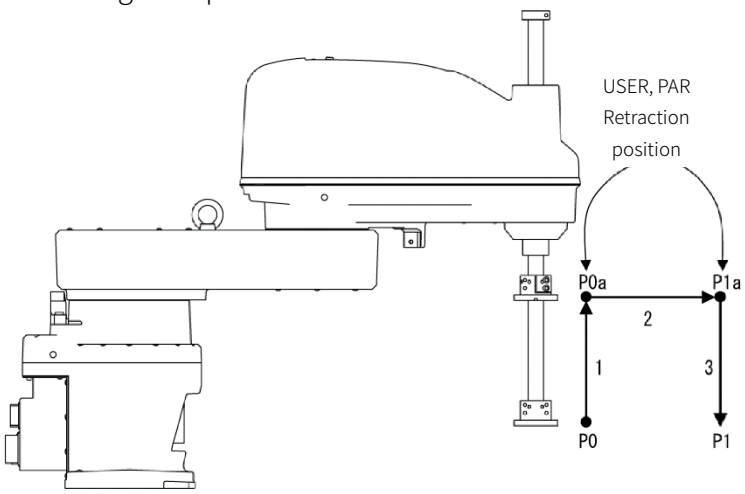


Figure 3.1 SCARA robot BYPASS

Parameter key	BYPASS-SW	
Parameter data	=(Function selected)	
Data type	Integer type	
Number of data items	1	
Data range	0 to 2	
Description	=0	Disabled
	=1	Enabled for MOVE TO and BYPASS.
	=2	Enabled for BYPASS only.
Default	=0	

Parameter key	BYPASS-RETRACT-POS
Parameter data	= (Axis 1), (Axis 2), (Axis 3), (Axis 4), (Axis 5), (Axis 6), (Axis 7), (Axis 8)
Data type	Real number type
Number of data items	8
Data range	-9999.999 to 9999.999(mm)
Description	Specify [Retraction position] by an absolute value. *1 Only axis 3 is enabled in SCARA robot.
Default	= 0.0, 0.0, 0.0, 0.0, 0.0, 0.0, 0.0, 0.0

Parameter key	BYPASS-ZOFST
Parameter data	= (System reserved 1), (System reserved 2)
Data type	Real number type
Number of data items	2
Data range	-
Description	System reserved.
Default	= 0.0, 0.0

Setting example

```
‘[U011] BYPASS FUNCTION
BYPASS-SW = 1
BYPASS-RETRACT-POS = 0.0, 0.0, 200.0, 0.0, 0.0, 0.0, 0.0, 0.0
BYPASS-ZOFST = 0.0, 0.0
```

NOTICE

- Before performing teach point movement, confirm whether the target position is correctly set.
  The robot that has made an unexpected motion may collide with a peripheral device.

[U012] COMMON DATA & PROGRAM FILE FUNCTION

Make program part and data part common

Specify the functions that make the program part and data part common for the program written in SCOL language.
The program written in SCOL language is divided into the program part where action and processing are written and the data part that has coordinate data such as POINT type data.

When the data part is made common, the program file (DATA.TBL) that has one data part can be used by multiple programs that have the data part in common. The program file that has the data part must be named DATA.TBL.

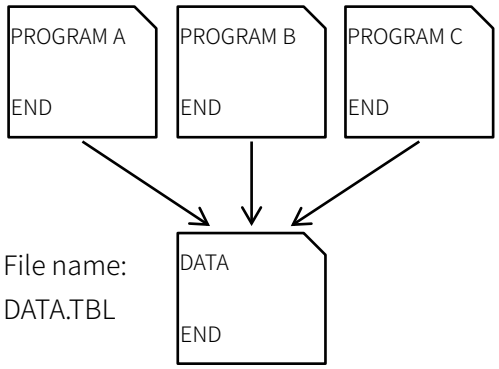


Figure 3.2 Making data part common

When the program part is made common, the program file (SCOL.PRG) that has one program part can be used by multiple programs in common. The program file that has the program part must be named SCOL.PRG.

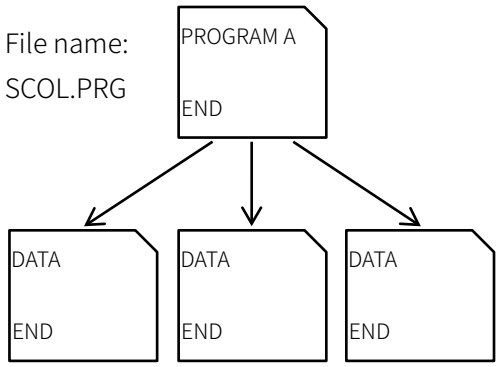


Figure 3.3 Making program part common

Parameter key	COMMON-FILE-FUNC	
Parameter data	=(Function selected)	
Data type	Integer type	
Number of data items	1	
Data range	0 to 2	
Description	=0	Disabled.
	=1	Make the data part common. “DATA.TBL” must be created.
	=2	Make the program part common. “SCOL.PRG” must be created.
Default	=0	

Setting example

```
‘[U012] COMMON DATA & PROGRAM FILE FUNCTION
COMMON-FILE-FUNC = 1
```

MEMO

・Applying the contents requires the power to be turned OFF and then turned ON.

[U013] PRINT ALIGNMENT

PRINT output zero suppress

Output the display output by the PRINT command as a decimal number of up to 10 digits when the value of the expression is integer type. Output it as 4 digits for integer part and 3 digits for fractional part (up to 7 digits) (8 digits including decimal point) when the value is real number type. The value is prefixed with a sign by one character, but the positive sign is omitted.
When the function is disabled (=0), the numeric value is right-justified and the remaining character part is padded with space.
When the function is enabled (=1), the display output by the PRINT command is set to be left-justified.

Example of output display

When disabled	□□□１００．０００
When enabled	１００．０００

Parameter key	PRINT-ALIGNMENT-SW	
Parameter data	=(Disabled/Enabled)	
Data type	Integer type	
Number of data items	1	
Data range	0 or 1	
Description	=0	Output 12 characters with blank padded with space.
	=1	Erase space and left-justify data.
Default	=1	

Setting example

```
‘[U013] PRINT ALIGNMENT
PRINT-ALIGNMENT-SW = 1
```

MEMO

For details about the PRINT command, refer to STA1163_ROBOT LANGUAGE MANUAL (SCOL2).

This function is enabled by default.

[U014] INPUT ALARM RESTART FUNCTION SW

Change INPUT error processing

When this function is enabled (=1),
an alarm (002-046 Channel error or 002-034 Ethernet connection error) occurs. If the program is re-executed after the alarm is cancelled, the program is executed from the line next to the interrupted line.
When this function is disabled (=0), an alarm caused by the program occurs. If the alarm is cancelled after the robot stops and the program is re-executed, the program is executed from the first line to perform program reset.

Parameter key	INPUT-RESTART-SW	
Parameter data	=(Disabled/Enabled)	
Data type	Integer type	
Number of data items	1	
Data range	0 or 1	
Description	=0	Program reset occurs and the program is executed from the beginning.
	=1	When the program is re-executed, the program is executed from the line next to the interrupted line.
Default	=0	

Setting example

```
'[U014] INPUT ALARM RESTART FUNCTION SW
INPUT-RESTART-SW = 1
```

MEMO

For details about the alarm, refer to STA1165_ALARM MANUAL.

[U015] DISABLE INPUT DATA WHILE PROGRAM STOP

Store INPUT buffer

Specify whether or not to empty the INPUT data buffer while the program is stopped.
When this function is disabled (=0), INPUT data is read and discarded and the buffer is emptied other than during program execution.
When this function is enabled (=1), the buffer is not emptied other than during program execution.

Parameter key	INPUT-BUFFER-CLEAR-SW	
Parameter data	=(Disabled/Enabled)	
Data type	Integer type	
Number of data items	1	
Data range	0 or 1	
Description	=0	The communication buffer is cleared while the program is stopped.
	=1	The communication buffer is not cleared even while the program is stopped..
Default	=0	

Setting example

```
‘[U015] DISABLE INPUT DATA WHILE PROGRAM STOP
INPUT-BUFFER-CLEAR-SW = 1
```

[U016] EDIT LOCK

Lock EDIT function

Specify a password to lock the EDIT function.
This function can be used on TP1000 only.

Parameter key	EDIT-LOCK-FUNCTION-SW	
Parameter data	=(Disabled/Enabled)	
Data type	Integer type	
Number of data items	1	
Data range	0 or 1	
Description	=0	EDIT lock is disabled.
	=1	EDIT lock is enabled.
Default	=0	

Parameter key	PASSWORD
Parameter data	=(PASSWORD)
Data type	Integer type
Number of data items	1
Data range	0 or 1
Description	Specify a password to unlock the EDIT function. Specify the password by up to 8 characters.
Default	" "

Setting example

```
‘[U016] EDIT LOCK
EDIT-LOCK-FUNCTION-SW = 1
PASSWORD = "PASSWORD"
```

MEMO

・Applying the contents requires the power to be turned OFF and then turned ON.

[U017] PASS COMMAND NO WAIT MODE

Switch PASS operation mode

Specify whether to perform PASS operation after completion of PASS path generation or without waiting for completion of PASS path generation.
When this function is disabled, the PASS operation is performed, but a waiting time for PASS operation generation may occur before the PASS command starts if the processing time of the program between PASS commands is long.
When this function is enabled, the PASS operation immediately after WAIT MOTION≧100 command can no longer be performed by the program.

Parameter key	PASS-CMD-MODE-SW	
Parameter data	=(Disabled/Enabled)	
Data type	Integer type	
Number of data items	1	
Data range	0 or 1	
Description	=0	(Disabled) Perform the PASS operation after completion of PASS path generation.
	=1	(Enabled) Perform the PASS operation without waiting for completion of PASS path generation.
Default	=0	

Setting example

```
‘[U017] PASS COMMAND NO WAIT MODE
PASS-CMD-MODE-SW = 0
```

[U018] AUX SIGNAL

Set auxiliary signal display (AUX)

Specify the data on the auxiliary signal display so that the robot I/O can be set ON and OFF from the teach pendant.
Single solenoid and double solenoid are supported.

Parameter key	AUX-SIG[num[1)]]
Parameter data	=(Signal name), (1st signal No.), (2nd signal No.)
Data type	String type
Number of data items	3×20
Data range	(Signal name) Up to 10 characters (Signal No.) Standard output: 1(DOUT1) to 64(DOUT64) Extended output: 101(DOUT101) to 164(DOUT164) Hand output: 201(HANDDOUT1) to 216(HANDDOUT16) Field bus: 301 to 364, 401 to 464, 501 to 564, 601 to 664, 701 to 764, 801 to 864, 901 to 964, 1001 to 1064

1) Num takes an integer from 0 to 19.

(Signal name)
Specify the signal name to be displayed.

(Signal No.)

Single solenoid

Perform normal output.
Specify the signal to be used in (1st signal No.).
Specify 0 in (2nd signal No.).

Double solenoid

Output two (2) output signals exclusively.
Specify the signal to be used in (1st signal No.), (2nd signal No.).

Setting example

Single solenoid

```
AUX-SIG[0] =”Hand out 1”, 201, 0
```

Register output signal 201 as single solenoid and specify the display name as Hand out 1.

Double solenoid

```
AUX-SIG[1] =”Hand out 2”, 201, 202
```

Register output signal 201, 202 as double solenoid and specify the display name as Hand out 2.

```
‘[U018] AUX SIGNAL
AUX-SIG[0] = ”Hand out 1”, 201, 0
AUX-SIG[1] = ”Hand out 2”, 202, 0
AUX-SIG[2] = ”Hand out 3”, 203, 0
AUX-SIG[3] = ”Hand out 4”, 204, 0
AUX-SIG[4] = ”Not Use”,0, 0
AUX-SIG[5] = ”Dout1”, 1, 0
AUX-SIG[6] = ”Dout2”, 2, 0
AUX-SIG[7] = ”Dout3”, 3, 0
AUX-SIG[8] = ”Dout4”, 4, 0
AUX-SIG[9] = ”Dout5”, 5, 0
AUX-SIG[10] = ”Dout6”, 6, 0
AUX-SIG[11] = ”Dout7”, 7, 0
AUX-SIG[12] = ”Dout8”, 8, 0
AUX-SIG[13] = ”Dout9”, 9, 0
AUX-SIG[14] = ”Dout10”, 10, 0
AUX-SIG[15] = ”Dout11”, 11, 0
AUX-SIG[16] = ”Dout12”, 12, 0
AUX-SIG[17] = ”Dout13”, 13, 0
AUX-SIG[18] = ”Dout14”, 14, 0
AUX-SIG[19] = ”Dout15”, 15, 0
```

[U019] RESTRICT SIGNAL OUTPUT

Restrict I/O, AUX

Prohibit the changing of I/O signals on the I/O and AUX screens in auto mode.
When this function is enabled (=1), I/O signals cannot be changed on the teach pendant in auto mode.

Parameter key	SIG-OUTPUT-PERMIT-SW	
Parameter data	=(Disabled/Enabled)	
Data type	Integer type	
Number of data items	1	
Data range	0 or 1	
Description	=0	Disable this function.
	=1	Prohibit the editing of I/O on the teach pendant in auto mode.
Default	=0	

Setting example

```
‘[U019] RESTRICT SIGNAL OUTPUT
SIG-OUTPUT-PERMIT-SW = 1
```

MEMO

Even if this function is enabled, I/O can be changed using PLC.

[U020] MANUAL MODE OVERRIDE LIMIT

Upper limit MANUAL mode override

Specify override in MANUAL mode.

Override is the motion speed when the maximum speed of motion is set as 100%.

Parameter key	TESTRUN-SPEED
Parameter data	=(OVERRIDE)
Data type	Integer type
Number of data items	1
Data range	1 to 100(%)
Default	=20

Setting example

```
‘[U020] MANUAL MODE OVERRIDE LIMIT
TESTRUN-SPEED = 20
```

MEMO

- *If override is set to 0, operation is performed at the speed specified in the internal parameter which was set at shipment.*
- *Applying the contents requires the power to be turned OFF and then turned ON.*

[U021] OVERRIDE LIMIT ON SLOW SPEED SIGNAL

Upper limit low-speed command override

Specify low-speed command (system input signal) override.
Override is the motion speed when the maximum speed of motion is set as 100%.
Specify at how many percent of the maximum speed the robot operates when a low-speed command signal is input during automatic operation.

Parameter key	SLOW-SPEED
Parameter data	=(OVERRIDE)
Data type	Integer type
Number of data items	1
Data range	1 to 100(%)
Default	=25

Setting example

```
‘[U021] OVERRIDE LIMIT ON SLOW SPEED SIGNAL
SLOW-SPEED= 25
```

MEMO

If the value is set to 0, the operation is performed at the speed specified in the internal parameter which was set at shipment.

[U022] OVERRIDE RETENTION SW

Override retention function in case of power failure

Retain the override.
When this function is disabled (=0), the override setting is initialized when the controller power is turned ON.
When this function is enabled (=1), the override retained when the controller power is shut OFF and then turned ON again is enabled.

Parameter key	OVRD-BACKUP-SW	
Parameter data	=(0/1)	
Data type	Integer type	
Number of data items	1	
Data range	0 or 1	
Description	=0	The override is initialized at power ON.
	=1	The override is not initialized at power ON. The override retained at power off is enabled.
Default	=0	

Setting example

```
‘[U022] OVERRIDE RETENTION SW
OVRD-BACKUP-SW = 1
```

MEMO

- *The override defaults to 20 in MANUAL mode and 100 in AUTO mode.*
- *Applying the contents requires the power to be turned OFF and then turned ON.*

[U023] BUZZER SILENT

Buzzer mute function

Set the buzzer when an alarm occurs.

Parameter key	BUZZER-SILENT-SW	
Parameter data	=(0/1)	
Data type	Integer type	
Number of data items	1	
Data range	0 or 1	
Description	=0	The buzzer is sounded when an alarm occurs.
	=1	The buzzer is not sounded when an alarm occurs.
Default	=0	

Setting example

```
‘[U023] BUZZER SILENT
BUZZER-SILENT-SW = 1
```

MEMO

When this function is enabled, the occurrence of an alarm cannot be confirmed using the buzzer sound.

[U024] NOT RECORD SAFETY SWITCH ALARM

8-017 History setting

Specify whether or not to record the 8-017 alarm on the error history.

Parameter key	SAFETY-SW-ALARM-RECORD-SW	
Parameter data	=(0/1)	
Data type	Integer type	
Number of data items	1	
Data range	0 or 1	
Description	=0	Recorded on the alarm history.
	=1	Not recorded on the alarm history.
Default	=1	

Setting example

```
‘[U024] NOT RECORD SAFETY SWITCH ALARM
SAFETY-SW-ALARM-RECORD-SW = 1
```

[U025] NOT RECORD EMERGENCY SWITCH ALARM

8-014 History setting

Specify whether or not to record the 8-014 alarm on the error history.

Parameter key	EMG-SW-ALARM-RECORD-SW	
Parameter data	=(0/1)	
Data type	Integer type	
Number of data items	1	
Data range	0 or 1	
Description	=0	Recorded on the alarm history.
	=1	Not recorded on the alarm history.
Default	=0	

Setting example

```
‘[U025] NOT RECORD EMERGENCY SWITCH ALARM
EMG-SW-ALARM-RECORD-SW = 1
```

[U026] USER ALARM

USER alarm message

Specify the message when USER alarm occurs.
The USER alarm can be generated by the ALARM command using the SCOL2 language.

The ALARM command is used in the program as follows.
ALARM(<alarm-level>, <alarm-number>)
For details about how to use the ALARM command, refer to STA1163_ROBOT LANGUAGE MANUAL (SCOL2).

Parameter key	USER-ALARM[num[1)]]
Parameter data	=(Alarm message)
Data type	String type
Number of data items	32
Data range	Up to 32 alphanumeric characters

1) num takes an integer from 0 to 31.

The correspondence between <alarm level>, <alarm number> and parameter key specified in the ALARM command is as follows.

<Alarm level>	<Alarm number>	Alarm	Parameter key
8	1	008-129	USER-ALARM[0]
	2	008-130	USER-ALARM[1]
	3	008-131	USER-ALARM[2]
	4	008-132	USER-ALARM[3]
	5	008-133	USER-ALARM[4]
	6	008-134	USER-ALARM[5]
	7	008-135	USER-ALARM[6]
	8	008-136	USER-ALARM[7]
4	1	004-129	USER-ALARM[8]
	2	004-130	USER-ALARM[9]
	3	004-131	USER-ALARM[10]
	4	004-132	USER-ALARM[11]
	5	004-133	USER-ALARM[12]
	6	004-134	USER-ALARM[13]
	7	004-135	USER-ALARM[14]
	8	004-136	USER-ALARM[15]

2	1	002-129	USER-ALARM[16]
	2	002-130	USER-ALARM[17]
	3	002-131	USER-ALARM[18]
	4	002-132	USER-ALARM[19]
	5	002-133	USER-ALARM[20]
	6	002-134	USER-ALARM[21]
	7	004-135	USER-ALARM[22]
	8	004-136	USER-ALARM[23]
1	1	002-129	USER-ALARM[24]
	2	002-130	USER-ALARM[25]
	3	002-131	USER-ALARM[26]
	4	002-132	USER-ALARM[27]
	5	002-133	USER-ALARM[28]
	6	002-134	USER-ALARM[29]
	7	004-135	USER-ALARM[30]
	8	004-136	USER-ALARM[31]

Setting example

‘[U026] USER ALARM
‘{008-129}
USER-ALARM[0]= “User Alarm 01”
‘{008-130}
USER-ALARM[1]= “User Alarm 02”
‘{008-131}
USER-ALARM[2]= “User Alarm 03”
•
•
•
‘{001-134}
USER-ALARM[29]= “User Alarm 30”
‘{001-135}
USER-ALARM[30]= “User Alarm 31”
‘{001-136}
USER-ALARM[31]= “User Alarm 32”

[U027] LOAD CELL SETTING

Load cell setting

Specify the load cell to be used.

When using this function, an optional A/D conversion board must be installed in the controller.

Parameter key	LOADCELL-USE-FLAG	
Parameter data	=(Disabled/Enabled)	
Data type	Integer type	
Number of data items	1	
Data range	0 or 1	
Description	=0	The load cell is not used.
	=1	The load cell is used.
Default	=0	

Parameter key	LOADCELL-RATED-CAPACITY
Parameter data	=(Rated capacity [N] of load cell)
Data type	Real number type
Number of data items	1
Data range	0.0 -
Description	Specify the rated capacity [N] of the load cell to be used.
Default	=0.0

Parameter key	LOADCELL-OUTPUT-VOLTAGE
Parameter data	=(Rated output [mV/V] of load cell)
Data type	Real number type
Number of data items	1
Data range	0.0 -
Description	Specify the rated output [mV/V] of the load cell to be used.
Default	=0.0

Parameter key	LOADCELL-DIRECTION	
Parameter data	=(+direction/-direction)	
Data type	Integer type	
Number of data items	1	
Data range	0 or 1	
Description	=0	A positive voltage is output when a load cell is imposed.
	=1	A negative voltage is output when a load cell is imposed.
Default	=0	

Parameter key	LOADCELL-ZERO-OFFSET
Parameter data	=(Zero-point offset)
Data type	Integer type
Number of data items	1
Data range	-32768 to 32767
Description	Change the zero point of the load cell.
Default	=0

Setting example

```
'[U027] LOAD CELL SETTING
'{0:Disabled 1:Enabled}
LOADCELL-USE-FLAG = 1
'{Rated Capacity [N]}
LOADCELL-RATED-CAPACITY = 245.2
'{Rated Output [mV/V]}
LOADCELL-OUTPUT-VOLTAGE = 1.0
'{Output Direction 0:+ 1:-}
LOADCELL-DIRECTION = 0
'{Zero Point Offset}
LOADCELL-ZERO-OFFSET = -382
```

[U028] SYSTEM VARIABLE ASYNC FLAG

Set asynchronous system variable access

When this flag is enabled, a system variable is accessed in SCOL language without waiting for completion of the motion command. This function is compatible with the TS3000 series and X8KCS-03B and previous versions. In other cases, it is recommended to disable this flag.

Parameter key	SYSTEM-VAR-ASYNC	
Parameter data	=(Disabled/Enabled)	
Data type	Integer type	
Number of data items	1	
Data range	0 or 1	
Description	=0	Completion of motion command is waited for when system variable is accessed.
	=1	Completion of motion command is not waited for when system variable is accessed.
Default	=0	
System version	X8KCS-03C and later	

Setting example

```
'[U028] SYSTEM VARIABLE ASYNC FLAG
'{0:Disabled 1:Enabled}
SYSTEM-VAR-ASYNC = 0
```

[U029] PRINT/INPUT CONTROL CODE CONVERSION

Setting control character conversion with PRINT/INPUT

This parameter sets whether or not to convert control characters for input/output when the PRINT/INPUT command is executed. When conversion is performed, the PRINT command converts the two-digit hexadecimal number enclosed by angle brackets (< >) into the corresponding ASCII code character and outputs it. When the INPUT command receives data with ASCII codes in the range of 0x00 to 0x1F and 0x7F and above, it converts data into a two-digit hexadecimal string enclosed by angle brackets (< >), and holds it in a string-type variable. For details, refer to the COMMUNICATION MANUAL.

Parameter key	CONTROL-CODE-CONVERSION	
Parameter data	=(Disabled/Enabled)	
Data type	Integer type	
Number of data items	1	
Data range	0 or 1	
Description	=0	No control character conversion is performed.
	=1	Control character conversion is performed.
Default	=0	
System version	X8KCS-05B and later	

Setting example

```
'[U029] PRINT/INPUT CONTROL CODE CONVERSION
'{0:Disabled 1:Enabled}
CONTROL-CODE-CONVERSION = 0
```

ETHERNET.PAR

This section describes ETHERNET.PAR.

ETHERNET.PAR sets Ethernet communication.

Performing Ethernet communication requires setting to be correctly performed.

For details, refer to STA1164_Communication Manual.

[E000] ROBOT CONTROLLER IP ADDRESS

Controller IP address

Set the IP address of the controller.

Parameter key	IP-ADDRESS
Parameter data	=(xxxx), (xxxx), (xxxx), (xxxx)
Data type	Integer type
Number of data items	4
Data range	-
Default	=192, 168, 0, 124

Setting example

```
'[E000] ROBOT CONTROLLER IP ADDRESS
IP-ADDRESS = 192, 168, 0, 124
```

MEMO

- *The numerical values are separated by a comma "," instead of a dot ".".*
- *Be careful not to cause the IP address in a network to overlap.*
- *Unless the IP address is correctly set, connection with the controller cannot be made.*
- *IP addresses "223.255.0.0" to "223.255.255.255" are reserved as the system by the controller. When setting the IP address of the controller, set the IP address that is not system-reserved.*
- *Applying the contents requires the power to be turned OFF and then turned ON.*

[E001] SUBNET MASK

Set controller subnet mask

Specify the subnet mask of the controller.

Parameter key	SUBNETMASK
Parameter data	=(xxxx), (xxxx), (xxxx), (xxxx)
Data type	Integer type
Number of data items	4
Data range	-
Default	=255, 255, 255, 0

Setting example

```
'[E001] SUBNET MASK
SUBNETMASK = 255, 255, 255, 0
```

MEMO

- *Unless the subnet mask is correctly set, connection with the controller cannot be made.*
- *Applying the contents requires the power to be turned OFF and then turned ON.*

[E002] DEFAULT GATEWAY

Set default gateway of controller

Set the default gateway of the controller.

Parameter key	GATEWAY
Parameter data	=(xxxx), (xxxx), (xxxx), (xxxx)
Data type	Integer type
Number of data items	4
Data range	-
Default	=192, 168, 0, 1

Setting example

```
'[E002] DEFAULT GATEWAY
GATEWAY = 192, 168, 0, 1
```

MEMO

- *Unless the default gateway is correctly set, connection with the controller cannot be made.*
- *Applying the contents requires the power to be turned OFF and then turned ON.*

[E003] ROBOT COMMUNITY NAME

Controller name

Set the name of the controller.

Parameter key	ROBO-COMMUNITY-NAME
Parameter data	=”Name”
Data type	String type
Number of data items	1
Data range	-
Default	=”TS5K-[Serial number]”

Setting example

```
‘[E003] ROBOT COMMUNITY NAME
ROBO-COMMUNITY-NAME = "TS5K"
```

MEMO

- *Set the robot controller name to a value that is unique on the same network.*
- *Applying the contents requires the power to be turned OFF and then turned ON.*

[E004] OPEN MODE IP1 IP2 IP3 IP4 IP5 IP6 IP7 IP8

IP open mode

Select the mode from IP1 to IP8.

Parameter key	OPEN-MODE	
Parameter data	=(IP1), (IP2), (IP3), (IP4), (IP5), (IP6), (IP7), (IP8)	
Data type	Integer type	
Number of data items	1	
Data range	0 to 3	
Description	=0	Unused
	=1	Set the controller to TCP/IP server and perform communication.
	=2	Set the controller to TCP/IP client and perform communication.
	=3	The controller uses UDP to perform communication.
Default	=1,0,0,0,0,0,0,0	

Setting example

```
‘[E004] OPEN MODE IP1 IP2 IP3 IP4 IP5 IP6 IP7 IP8
OPEN-MODE = 1, 2, 2, 0, 0, 0, 0, 0
```

MEMO

- *Unless the server, client and UDP are correctly set, connection with the controller cannot be made.*

[E005] OWN PORT NUMBER

Controller port number

Set the port numbers assigned to IP1 to IP8 of the controller.

Parameter key	OWN-PORT-NO[num[1)]]
Parameter data	=(Port number)
Data type	Integer type
Number of data items	1
Data range	10000 to 65535

1) num takes an integer from 0 to 7.

Setting example

```
‘[E005] OWN PORT NUMBER
‘{IP1}
OWN-PORT-NO[0] = 10000
‘{IP2}
OWN-PORT-NO[1] = 10001
‘{IP3}
OWN-PORT-NO[2] = 10002
‘{IP4}
OWN-PORT-NO[3] = 10003
‘{IP5}
OWN-PORT-NO[4] = 10004
‘{IP6}
OWN-PORT-NO[5] = 10005
‘{IP7}
OWN-PORT-NO[6] = 10006
‘{IP8}
OWN-PORT-NO[7] = 10007
```

MEMO

- *This function is used when OPEN MODE set in [E004] is server (=1).*
- *Set the port numbers so that they do not overlap.*
- *Unless the port numbers are correctly set, connection with the controller cannot be made.*

[E006] IP ADDRESS OF DESTINATION

Connection destination IP address

Set connection destination IP addresses from IP1 to IP8.

Parameter key	DEST-IP-ADR[num[1)]]
Parameter data	=(xxxx), (xxxx), (xxxx), (xxxx)
Data type	Integer type
Number of data items	4
Data range	-

1) num takes an integer from 0 to 7.

Setting example

```
‘[E006] IP ADDRESS OF DESTINATION
‘{IP1}
DEST-IP-ADR[0] = 192, 168, 0, 150
‘{IP2}
DEST-IP-ADR[1] = 192, 168, 0, 151
‘{IP3}
DEST-IP-ADR[2] = 192, 168, 0, 152
‘{IP4}
DEST-IP-ADR[3] = 192, 168, 0, 153
‘{IP5}
DEST-IP-ADR[4] = 192, 168, 0, 154
‘{IP6}
DEST-IP-ADR[5] = 192, 168, 0, 155
‘{IP7}
DEST-IP-ADR[6] = 192, 168, 0, 156
‘{IP8}
DEST-IP-ADR[7] = 192, 168, 0, 157
```

MEMO

- *This function is used when OPEN MODE set in [E004] is client (=2).*
- *Unless the IP addresses are correctly set, connection with the controller cannot be made.*

[E007] PORT NUMBER OF DESTINATION

Connection destination port number

Set the connection destination port numbers from IP1 to IP8.

Parameter key	DEST-PORT-NO [num[1)]]
Parameter data	=(Connection destination port number)
Data type	Integer type
Number of data items	4
Data range	-

1) num takes an integer from 0 to 7.

Setting example

```
‘[E007] PORT NUMBER OF DESTINATION
‘{IP1}
DEST-PORT-NO[0] = 10000
‘{IP2}
DEST-PORT-NO[1] = 10001
 ‘{IP3}
DEST-PORT-NO[2] = 10002
 ‘{IP4}
DEST-PORT-NO[3] = 10003
 ‘{IP5}
DEST-PORT-NO[4] = 10004
 ‘{IP6}
DEST-PORT-NO[5] = 10005
 ‘{IP7}
DEST-PORT-NO[6] = 10006
 ‘{IP8}
DEST-PORT-NO[7] = 10007
```

MEMO

- *This function is used when OPEN MODE set in [E004] is client (=2).*
- *Unless the port numbers are correctly set, connection with the controller cannot be made.*

[E008] TCP STATUS ALARM IP1 IP2 IP3 IP4 IP5 IP6 IP7 IP8

TCP status alarm

Set the alarm level that is generated for controller IP1 to IP8 when Ethernet connection is closed while the program runs.

Parameter key	TCP-STATUS-ALM
Parameter data	=(IP1), (IP2), (IP3), (IP4), (IP5), (IP6), (IP7), (IP8)
Data type	Integer type
Number of data items	8
Data range	Any one of 0, 1, 2, 4, or 8
Default	=0, 0, 0, 0, 0, 0, 0, 0

Setting example

```
‘[E009] TCP STATUS ALARM IP1 IP2 IP3 IP4 IP5 IP6 IP7 IP8
TCP-STATUS-ALM = 8, 4, 2, 0, 0, 0, 0, 0
```

MEMO

If the value is set to 0 (zero), the alarm is not generated.

PLC.PAR

This section describes PLC.PAR.

PLC.PAR performs the setting related to TCmini (built-in PLC).

[P000] I/O MODE

I/O mode

Specify the reading destination and writing destination of the ladder program.
After setting, the power must be turned OFF and then turned ON.

Parameter key	IO-MODE	
Parameter data	=(Mode)	
Data type	Integer type	
Number of data items	1	
Data range	0 to 2	
Description	=0	Standard sequence area
	=1	Debug user RAM
	=2	User flash memory
Default	=0	

Setting example

```
‘[P000] IO-MODE = 0
IO-MODE = 0
```

MEMO

- *Data cannot be written into the standard sequence area.*
- *For details about change of the ladder program, refer to STA1168 TCmini (BUILT-IN PLC) MANUAL.*
- *Applying the contents requires the power to be turned OFF and then turned ON.*

[P001] DATA REGISTER HOLD

Retain data register

Retain the data register if I/O mode in [P000] is set to user flash memory (=2).

Save the values of data registers (from D300 to D37F) in the nonvolatile memory in the controller.

Parameter key	LATCH-REGISTER-HOLD-SW	
Parameter data	=(1/0)	
Data type	Integer type	
Number of data items	1	
Data range	0 or 1	
Description	=0	The latch register is not retained.
	=1	The latch register is retained.
Default	=0	

Setting example

```
'[P001] DATA REGISTER HOLD
DATA-REGISTER-HOLD-SW = 1
```

MEMO

- *Applying the contents requires the power to be turned OFF and then turned ON.*

[P002] SEQUENCE PARAMETER

Sequence parameter

Set the value of the internal relay to be used as contact input in the user-created ladder program.
This function is enabled when [P000] I/O MODE is set to 1 or 2.

Parameter key	SEQ-PARA
Parameter data	=(H110), (H111), (H112), (H113), (H114), (H115), (H116), (H117)
Data type	Integer type
Number of data items	1
Data range	0 or 1
Default	=0, 0, 0, 0, 0, 0, 0, 0

Setting example

```
'[P002] SEQUENCE PARAMETER
SEQ-PARA = 0, 0, 0, 0, 0, 0, 0, 0
```

MEMO

- *Applying the contents requires the power to be turned OFF and then turned ON.*

[P003] SEQUENCE FUNCTION SELECT SWITCH

Allocate system signal Fieldbus

Set the value of internal relay to be used as contact input in the user-created ladder program.
When using the standard ladder, fieldbus allocation is switched. For details, refer to STA1168_TCmini (Internal PLC) MANUAL.

Parameter key	SEQ-FUNCTION-SW
Parameter data	=(H118), (H119), (H11A), (H11B), (H11C), (H11D), (H11E), (H11F)
Data type	Integer type
Number of data items	8
Data range	0 or 1
Default	=0, 0, 0, 1, 0, 0, 0, 0

Setting example

```
‘[P003] SEQUENCE FUNCTION SELECT SWITCH
SEQ-FUNCTION-SW = 0, 0, 0, 1, 0, 0, 0, 0
```

MEMO

- *Applying the contents requires the power to be turned OFF and then turned ON.*

[P004] PLC ALARM

PLC alarm message

Set PLC alarm message.

PLC alarm can be generated from PLC when [P000] I/O mode setting is user.

Parameter key	USER-PLC-ALARM[num[1)]]
Parameter data	=(Alarm message)
Data type	String type
Number of data items	16
Data range	Up to 32 alphanumeric characters

1) num takes an integer from 0 to 15.

Setting example

```
‘[P004] PLC ALARM
‘{008-269}
USER-PLC-ALARM[0]= "PLC alarm 01"
‘{008-270}
USER-PLC-ALARM[1]= "PLC alarm 02"
‘{008-271}
USER-PLC-ALARM[2]= "PLC alarm 03"
        •
        •
        •
‘{001-042}
USER-PLC-ALARM[13]= "PLC alarm 14"
‘{001-043}
USER-PLC-ALARM[14]= "PLC alarm 15"
‘{001-044}
USER-PLC-ALARM[15]= "PLC alarm 16"
```

[P005] PLC OVERRIDE LIMIT

PLC override limit

Set upper-limit value and lower-limit value of PLC override.

Parameter key	PLC-OVRD-UPPER
Parameter data	=(Upper-limit value of override)
Data type	Integer type
Number of data items	1
Data range	0 to 100
Default	100

MEMO

・When a value exceeding 100 is set, the value is set to 100.

Parameter key	PLC-OVRD-LOWER
Parameter data	=(Lower-limit value of override)
Data type	Integer type
Number of data items	1
Data range	0 to 100
Default	1

MEMO

・When a value that is lower than 0 is set, the value is set to 0.

Setting example

```
‘[P005] PLC OVERRIDE LIMIT
PLC-OVRD-UPPER = 50
PLC-OVRD-LOWER = 5
```

[P006] SIGNAL FEEDHOLD FUNCTION

FEEDHOLD by TCmini (built-in PLC)

Specify whether (FEEDHOLD)/(FEEDHOLD cancellation) operation is possible or impossible from TCmini.
When this setting is enabled, FEEDHOLD/FEEDHOLD cancellation operation can be executed by TCmini register G12C.
During program execution, when G12C is OFF, FEEDHOLD state is entered, and when G12C is ON, FEEDHOLD is canceled.
For details, refer to STA1168_TCmini (BUILT-IN PLC) MANUAL.

Parameter key	SIG-FEEDHOLD-SW	
Parameter data	=(Operation possible/impossible)	
Data type	Integer type	
Number of data items	1	
Data range	0 or 1	
Description	=0	Disabled (FEEDHOLD/FEEDHOLD cancellation from TCmini is impossible.)
	=1	Enabled (FEEDHOLD/FEEDHOLD cancellation from TCmini is possible.)
Default	=0	

Setting example

```
'[P006] SIGNAL FEEDHOLD FUNCTION
SIG-FEEDHOLD-SW = 1
```

[P007] FUNCTION WHICH TURNS ON AN ALARM SIGNAL TO ALL ALARMS

Set output alarm signal

Change the alarm level at which the output alarm signal (H13D) is turned on.
By default, the signal is turned on due to level 2, 4, and 8 alarms.
When this parameter is set to (=1) or (=2), the output alarm signal can be output due to level 1 alarm.

Parameter key	ALL-ALARM-SIG-FLG	
Parameter data	=(Alarm level selected)	
Data type	Integer type	
Number of data items	1	
Data range	0 to 2	
Description	=0	The signal is turned on due to level 2, 4, and 8 alarms.
	=1	The signal is turned on due to all level alarms (1, 2, 4, and 8).
	=2	The signal is turned on due to alarms other than emergency stop.
Default	=0	

Setting example

```
'[P007] FUNCTION WHICH TURNS ON AN ALARM SIGNAL TO ALL ALARMS
ALL-ALARM-SIG-FLG = 0
```

MEMO

(H13D) is the interface relay that indicates the occurrence of level 2 and 4 faults or level 8 fault in the robot controller and robot body. For details, refer to STA1168_TCmini (BUILT-IN PLC) MANUAL.

[P008] Reserve

Reserved

P008 is a reserved parameter.

[P009] FIELDBUS COMMON SETTING

For details about fieldbus, refer to STA1190_FIELDBUS SLAVE MANUAL.

Parameter key	FBUS-ENABLE
Parameter data	= (Fieldbus communication)
Data type	Integer type
Number of data items	1
Data range	-
Description	Enable/disable fieldbus communication. When disabled, fieldbus communication is not performed.
	0: Disable fieldbus communication. 1: Enable fieldbus communication.
Default	1

Setting example

: Disable fieldbus communication

```
‘[P009] FIELDBUS COMMON SETTING
FBUS-ENABLE = 0
```

Parameter key	FBUS-ERROR-JOG-ENABLE
Parameter data	= (Fieldbus alarm enabled/disabled)
Data type	Integer type
Number of data items	1
Data range	-
Description	This parameter temporarily disables the fieldbus alarm because the servo power cannot be turned on/off when an alarm is detected in fieldbus setting. * The function that disables the alarm using this parameter is only enabled in MANUAL mode. In AUTO mode, the alarm is not disabled even if this parameter is set to 1.
	0: Fieldbus alarm is displayed. 1: Alarm is disabled.
Default	0

Setting example

: Disable fieldbus alarm in MANUAL mode

```
‘[P009] FIELDBUS COMMON SETTING
FBUS-ERROR-JOG-ENABLE = 1
```

Parameter key	FIELDBUS-ENDIAN
Parameter data	= (Fieldbus endian)
Data type	Integer type
Number of data items	1
Data range	-
Description	Set the endian format of the fieldbus master device. The controller communicates with the fieldbus master by converting data into a bit field that corresponds to the endian format of the fieldbus master.
	0 ：16bit big endian (0x1234 5678→0x3412 7856) 1 ：16bit little endian (0x1234 5678→0x1234 5678) 2 ：32bit big endian (0x1234 5678→0x7856 3412) 3 ：32bit little endian (0x1234 5678→0x5678 1234)
Default	1

Setting example

: When the fieldbus master device is 16-bit little-endian

‘[P009] FIELDBUS COMMON SETTING FIELDBUS-ENDIAN = 1

[P010] ETHERNET/IP SETTING

For details about fieldbus, refer to STA1190_FIELDBUS SLAVE MANUAL.

Parameter key	ETHERNETIP-ADD
Parameter data	= (IP address)
Data type	Integer type
Number of data items	4
Data range	-
Description	• An IP address is like an address for identifying EtherNet connected devices. • An IP address consists of four numbers from 0 to 255 with comma (,) in between. • When connecting EtherNet/IP and a computer connected to the network directly on a one-to-one basis, the initial value (any value) for the IP address can be used; however, when connecting to an existing LAN, make an IP address setting according to the instructions of your LAN system administrator. If an IP address duplication occurs, the function will not work properly. • The computer or PLC that communicates with EtherNet/IP must be connected on the same network.
Default	= 192,168,0,1

Parameter key	ETHERNETIP-MASK
Parameter data	= (Subnet mask)
Data type	Integer type
Number of data items	4
Data range	-
Description	Specify the subnet mask of the network to be connected. When connecting to an existing LAN, make this setting of the subnet mask according to the instructions of your LAN system administrator.
Default	=255,255,255,0

Parameter key	ETHERNETIP-GATEWAY
Parameter data	= (Gateway)
Data type	Integer type
Number of data items	4
Data range	-
Description	Specify the default gateway address. When connecting to an existing LAN, make this setting of the default gateway address according to the instructions of your LAN system administrator.
Default	=0,0,0,0

Setting example

: Select EtherNet/IP and make the following setting for Keyence KV-7500.

IP address: 192.168.10.1

Subnet mask: 255.255.255.0

Gateway: 0.0.0.0

```
'[P010] ETHERNET/IP SETTING
'{IPAdress}
ETHERNETIP-ADD = 192, 168, 10, 10
'{Net Mask}
ETHERNETIP-MASK = 255, 255, 255, 0
'{GateWay}
ETHERNETIP-GATEWAY = 0, 0, 0, 0
```

[P011] CC-Link SETTING

For details about fieldbus, refer to STA1190_FIELDBUS SLAVE MANUAL.

Parameter key	CCLINK-NODE
Parameter data	= (Node)
Data type	Integer type
Number of data items	1
Data range	0 to 6 1
Description	Specify the node address of the fieldbus. Specify the address so that it does not overlap with other devices in the same network. The range of the setting value varies depending on the type of the fieldbus.
	0: No fieldbus option 1 to 64: Effective range of node address for CC-Link
Default	1

Parameter key	CCLINK-SPEED
Parameter data	= (Communication speed)
Data type	Integer type
Number of data items	1
Data range	0 to 4
Description	Specify the communication speed according to the communication speed of the master.
	0: 156 kbps 1: 625 kbps 2: 2.5 Mbps 3: 5 Mbps 4: 10 Mbps
Default	0

Parameter key	CCLINK-MODE
Parameter data	= (Number of inputs and outputs)
Data type	Integer type
Number of data items	1
Data range	0 to 2
Description	Specify the number of inputs and outputs used for CC-Link.
	=0 128 inputs and 128 outputs (16 bytes), 16 words (TS3000 compatible CC-LINK Ver. 1) =1 128 inputs and 128 outputs (16 bytes), 16 words (4-station, 1x, CC-LINK Ver.2) =2 224 inputs and 224 outputs (28 bytes), 32 words (4-station, 2x, CC-LINK Ver.2)
Default	0

Setting example

: Node address 1, communication speed 156 kbps, 224 inputs and 224 outputs (28 bytes), and 32 words

```
‘[P011]    CC-Link SETTING
‘{Node Addr}
‘{0:DISABLE, 1-255: ENABLE}
CCLINK-NODE = 1
‘{Speed}
CCLINK-SPEED = 0
‘{Mode}
CCLINK-MODE = 2
```

[P012] PROFIBUS SETTING

For details about fieldbus, refer to STA1190_FIELDBUS SLAVE MANUAL.

Parameter key	PROFIBUS SETTING	
Parameter data	= (Node)	
Data type	Integer type	
Number of data items	1	
Data range	0 to 125	
Description	(Node)	Specify the node address of the fieldbus. Make a setting so that it does not overlap with other devices in the same network. The range of the setting value varies depending on the type of the fieldbus.
Default	= 0	

Setting example

: Node address 2

```
‘[P012]    PROFIBUS SETTING
{ Node Addr}
{0-125 }
PROFIBUS-NODE = 2
```

[P013] DEVICENET SETTING

For details about fieldbus, refer to STA1190_FIELDBUS SLAVE MANUAL.

Parameter key	DEVICENET-NODE
Parameter data	= (Node)
Data type	Integer type
Number of data items	1
Data range	0 to 63
Description	Specify the node address of the fieldbus. Specify the address so that it does not overlap with other devices in the same network. The range of the setting value varies depending on the type of the fieldbus.
	1 to 63: Effective range of node address for DeviceNet
Default	1

Parameter key	DEVICENET-SPEED
Parameter data	= (Communication speed)
Data type	Integer type
Number of data items	1
Data range	0 to 2
Description	Specify the communication speed according to the communication speed of the master. No fieldbus option (initial value). 0: 125 kbps 1: 250 kbps 2: 500 kbps
Default	0

Setting example

: Node address 1 and communication speed 125 kbps

```
‘[P013]    DEVICENET SETTING
‘{Node Addr}
‘{0:DISABLE, 1-63: ENABLE}
DEVICENET-NODE = 1
‘{Speed}
DEVICENET-SPEED = 0
```

[P014] PROFINET SETTING

For details about fieldbus, refer to STA1190_FIELDBUS SLAVE MANUAL.

Parameter key	PROFINET-DCP-ENABLE
Parameter data	= (IP address specification with DCP)
Data type	Integer type
Number of data items	1
Data range	0 to 1
Description	0: Disable IP address specification with DCP. (IP address settings with a PLC.PAR file is enabled.) 1: Enable IP address specification with DCP. (IP address settings with a PLC.PAR file is disabled.)
Default	= 1

Parameter key	PROFINET-ADD
Parameter data	= (IP address)
Data type	Integer type
Number of data items	4
Data range	-
Description	• An IP address is like an address for identifying EtherNet connected devices. • An IP address consists of four numbers from 0 to 255 with comma (,) in between. • When connecting PROFINET and a computer connected to the network directly on a one-to-one basis, the initial value (any value) for the IP address can be used; however, when connecting to an existing LAN, make an IP address setting according to the instructions of your LAN system administrator. If an IP address duplication occurs, the function will not work properly. • The computer or PLC that communicates with PROFINET must be connected on the same network.
Default	= 192,168,0,1

Parameter key	PROFINET-MASK
Parameter data	= (Subnet mask)
Data type	Integer type
Number of data items	4
Data range	-
Description	Specify the subnet mask of the network to be connected. When connecting to an existing LAN, make this setting of the subnet mask according to the instructions of your LAN system administrator.
Default	=255,255,255,0

Parameter key	PROFINET -GATEWAY
Parameter data	= (Gateway)
Data type	Integer type
Number of data items	4
Data range	-
Description	Specify the default gateway address. When connecting to an existing LAN, make this setting of the default gateway address according to the instructions of your LAN system administrator.
Default	=0,0,0,0

Setting example

: Select PROFINET and make the following setting for Siemens AG S7-1200.

IP address: 192.168.0.10

Subnet mask: 255.255.255.0

Gateway: 0.0.0.0

```
'[P014] PROFINET SETTING
'{DCP Enable}
PROFINET-DCP-ENABLE = 1
'{IPAdress}
PROFINET -ADD = 192, 168, 0, 1
'{Net Mask}
PROFINET -MASK = 255, 255, 255, 0
'{GateWay}
PROFINET -GATEWAY = 0, 0, 0, 0
```

[P015] CLEAR PLC DATA AFTER PROGRAM RESET

Clear D register output after program reset

Specify whether to clear output to D register (D600-D63F) using PLCDATAW1 to 8, PLCSSW01 to 08, and PLCSLW01 to 08 to 0 after program reset or program select is done.

Parameter key	RESET-PLCDATAW	
Parameter data	= (0/1)	
Data type	Integer type	
Number of data items	1	
Data range	0 or 1	
Description	= 0	Hold the value after program reset.
	= 1	Clear the value after program reset.
Default	= 0	

Setting example

```
'[P015] CLEAR PLC DATA AFTER PROGRAM RESET
'{0:Disabled , 1:Enabled}
RESET-PLCDATAW = 0
```

[P016] MODBUS PROTCOL SELECT

This parameter specifies the protocol settings required to enable the Modbus TCP function.

For details about the Modbus TCP function, refer to STA1190_FIELDBUS SLAVE MANUAL.

Parameter key	MODOBUS-KIND
Parameter data	= (Modbus protocol type)
Data type	Integer type
Number of data items	1
Data range	0, 3
Description	0: Modbus function disabled 3: Modbus TCP Server function enabled
Default	= 0

Setting example

```
'[P016] CLEAR PLC DATA AFTER PROGRAM RESET
'{0:Disabled , 1:Enabled}
RESET-PLCDATAW = 0
```

EXTRNSEL.PAR

This section describes EXTRNSEL.PAR.

EXTRNSEL.PAR makes setting for selecting the program from the sequencer.
This function enables an execution file to be selected using the input signal.

[X000] SPECIFY SIGNAL FOR EXTSELECT

Allocate signal for EXTSELECT

Select an execution file using the input signal.
Allocate the signals used for EXTSELECT.

Parameter key	FILE-SEL-SIG
Parameter data	=(Signal No.), (bit length)
Data type	Integer type
Number of data items	2
Data range	-
Default	=1,0

(Signal No.)
Specify the leading number of the input signal to be used.
Specifiable number
Standard input, extended input

(bit length)
Specify the number of signals in the range from 1 to 8.

Setting example

```
‘[X000] SPECIFY SIGNAL FOR EXTSELECT
FILE-SEL-SIG = 1,8
```

MEMO

For details about signal timing and usage, refer to the section on STROBE of STA1168_TCmini (Internal PLC) MANUAL.

[X000-num] EXTRNSEL TABLE

TABLE

Specify the save destination and name of the program to be called using EXTRNSEL.
Up to 256 programs can be selected by setting the bit length in [X000] SPECIFY SIGNAL FOR EXTSELECT.

Parameter key	FILE[num[1)]].DRIVE	
Parameter data	=(Save destination)	
Data type	Integer type	
Number of data items	1	
Data range	0 or 1	
Description	=0	SD card
	=1	Controller
Default	=1	

Parameter key	FILE[num[1)]].NAME
Parameter data	=(Program name)
Data type	String type
Number of data items	1
Description	The program name is specified.
Default	=”PROG*”

1) num takes an integer from 0 to 255.

Setting example

```
'{Table No.0}
FILE[0].DRIVE = 1
FILE[0].NAME = "PROG000"
    •
    •
    •
'{Table No.255}
FILE[255].DRIVE= 0
FILE[255].NAME = "PROG255"
```

VISION.PAR

This section describes VISION.PAR.

VISION.PAR is used to calculate vision calibration and save the setting.

[V000] Coordinate transformation parameter for 2D vision calibration

3-point calibration

Specify the coordinate for performing the coordinate.
Input the set values in the following Table.

TABLE

Specify the coordinate for performing 3-point calibration.
Discrepancy between the coordinate viewed from the vision sensor and the coordinate viewed from the robot is corrected by setting the 3 points in the coordinate indicated by the robot for the 3 points in the coordinate indicated by the vision.
Up to 20 settings can be saved.

Parameter key	CALIB[num[1)]].VISION-REF
Parameter data	=(x1), (y1), (x2), (y2), (x3), (y3)
Data type	Real number type
Number of data items	6
Data range	-9999.999 to 9999.999
Description	Vision coordinate is set.

Parameter key	CALIB[num[1)]].ROBOT-REF
Parameter data	=(x1), (y1), (x2), (y2), (x3), (y3)
Data type	Real number type
Number of data items	6
Data range	-9999.999 to 9999.999
Description	Robot coordinate is set.

Parameter key	CALIB[num[1)]].METHOD
Parameter data	=(Type)
Data type	Integer type
Number of data items	1
Data range	0 only
Description	The type of calibration is set. Set the type to 0 to perform 3-point teaching.

Parameter key	CALIB[num[1)]].COORD	
Parameter data	=(Rotation direction)	
Data type	Integer type	
Number of data items	1	
Data range	0 to 3	
Description	=0	Right-handed (counterclockwise)
	=1	Right-handed (clockwise)
	=2	Left-handed (counterclockwise)
	=3	Left-handed (clockwise)

1) num takes an integer from 0 to 19.

Default value

```
'{Table No.1}
CALIB[0].VISION-REF = 0.000000, 0.000000, 0.000000, 0.000000, 0.000000, 0.000000
CALIB[0].ROBOT-REF = 0.000000, 0.000000, 0.000000, 0.000000, 0.000000, 0.000000
CALIB[0].METHOD = 0
CALIB[0].COORD = 0
    •
    •
    •
'{Table No.20}
CALIB[19].VISION-REF = 0.000000, 0.000000, 0.000000, 0.000000, 0.000000, 0.000000
CALIB[19].ROBOT-REF = 0.000000, 0.000000, 0.000000, 0.000000, 0.000000, 0.000000
CALIB[19].METHOD = 0
CALIB[19].COORD = 0
```

MEMO

The table No. is used in VSCALIBIN in SCOL2 language.

For details, refer to SCOL2.

[V001] Coordinate transformation matrix for multiple point calibration

9-point calibration for TSAssist

Save the set value obtained by performing 9-point calibration using SHIBAURA MACHINE integrated personal computer software TSAssist. The set value is input to the following table.

TABLE

This parameter saves the result of 9-point calibration using TSAssist
Up to 20 calculation results are saved.

Parameter key	NINE-CALIB[num[1)]].CALB-MTRX[0] NINE-CALIB[num[1)]].CALB-MTRX[1] NINE-CALIB[num[1)]].CALB-MTRX[2] NINE-CALIB[num[1)]].CALB-MTRX[3]
Parameter data	=(00), (01), (02), (03) =(10), (11), (12), (13) =(20), (21), (22), (23) =(30), (31), (32), (33)
Data type	Real number type
Number of data items	4×4
Data range	-9999.999 to 9999.999
Description	The result of 9-point calibration is saved.

Parameter key	NINE-CALIB[num[1)]].ROT-ANGL
Parameter data	=(Angle)
Data type	Real number type
Number of data items	1
Data range	-9999.999 to 9999.999(deg)
Description	The angle of 9-point calibration is saved.

Parameter key	CALIB[num[1)]].COORD	
Parameter data	=(Rotation direction)	
Data type	Integer type	
Number of data items	1	
Data range	0 or 1	
Description	=0	CCW
	=1	CW

1) num takes an integer from 0 to 19.

Default value

```
'{Table No.21}
NINE-CALIB[0].CALB-MTRX[0] = 0.000000, 0.000000, 0.000000, 0.000000
NINE-CALIB[0].CALB-MTRX[1] = 0.000000, 0.000000, 0.000000, 0.000000
NINE-CALIB[0].CALB-MTRX[2] = 0.000000, 0.000000, 0.000000, 0.000000
NINE-CALIB[0].CALB-MTRX[3] = 0.000000, 0.000000, 0.000000, 0.000000
NINE-CALIB[0].ROT-ANGL = 0.000000
NINE-CALIB[0].ROT-DRCT = 0
  •
  •
  •
'{Table No.40}
NINE-CALIB[19].CALB-MTRX[0] = 0.000000, 0.000000, 0.000000, 0.000000
NINE-CALIB[19].CALB-MTRX[1] = 0.000000, 0.000000, 0.000000, 0.000000
NINE-CALIB[19].CALB-MTRX[2] = 0.000000, 0.000000, 0.000000, 0.000000
NINE-CALIB[19].CALB-MTRX[3] = 0.000000, 0.000000, 0.000000, 0.000000
NINE-CALIB[19].ROT-ANGL = 0.000000
NINE-CALIB[19].ROT-DRCT = 0
```

MEMO

The TABLE No. is used in VSCALIBIN in SCOL2 language.

For details, refer to SCOL2.

AREA.PAR

This section describes AREA.PAR.

AREA.PAR enables the area to be set and linked with the output signal.

⚠ WARNING

- After setting the area, be sure to confirm that the specified area is correctly set.
- After setting the area, confirm the influence of the output signal when turning off the power to the controller.
- Confirm the influence of the output signal before the robot operates.
  The robot that has made an unexpected motion may collide with a peripheral device.

[A000] AREA OUTPUT SETTING PARAMETER

Area output setting

Link the output signal according to the position of TCP.
Set the area to be enabled or disabled, area size, and output signal.
Up to 64 area settings can be saved.

Area

Type 0: 2-point cuboid

Set the area using a cuboid created by specifying two points.
Set the coordinates of point 1 with AREA[num].DEFINITION1 and those of point 2 with AREA[num].DEFINITION2.

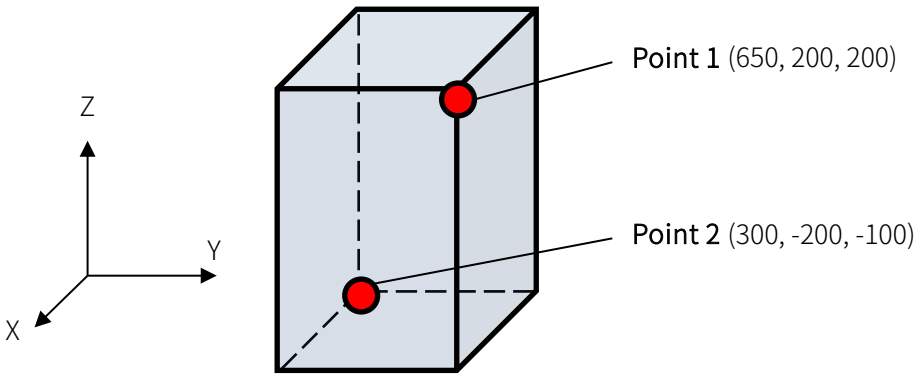


Figure 3.4 Example of setting a 2-point cuboid area

Type 1: Cuboid

Set the area using a cuboid created by specifying the center position, rotation, and shape dimensions.
Set the coordinates of the center position and rotation with AREA[num]. DEFINITION1 and the shape dimensions with AREA[num]. DEFINITION2.

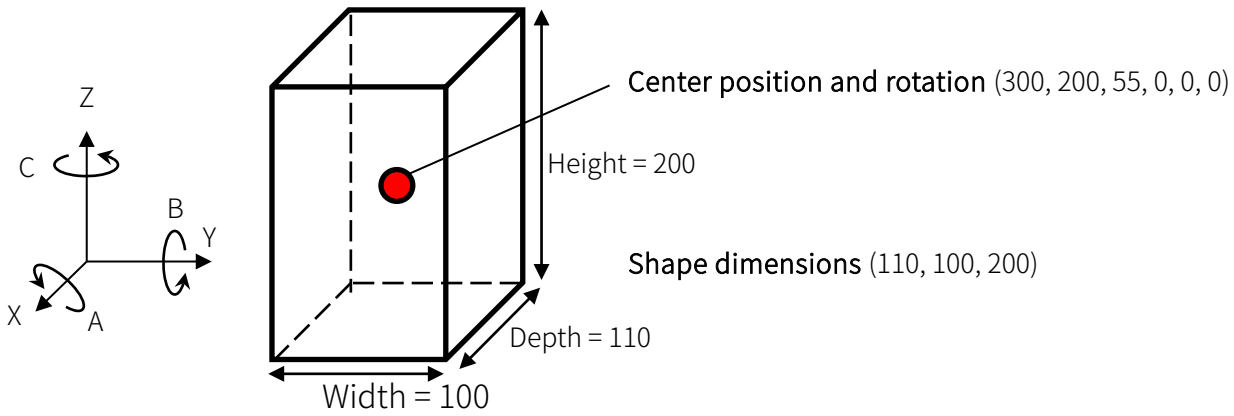


Figure 3.5 Example of setting a cuboid area

Type 2: Cylinder

Set the area using a cylinder created by specifying the center position, rotation, and shape dimensions.
Set the coordinates of the center position and rotation with AREA[num]. DEFINITION1 and the shape dimensions with AREA[num]. DEFINITION2.

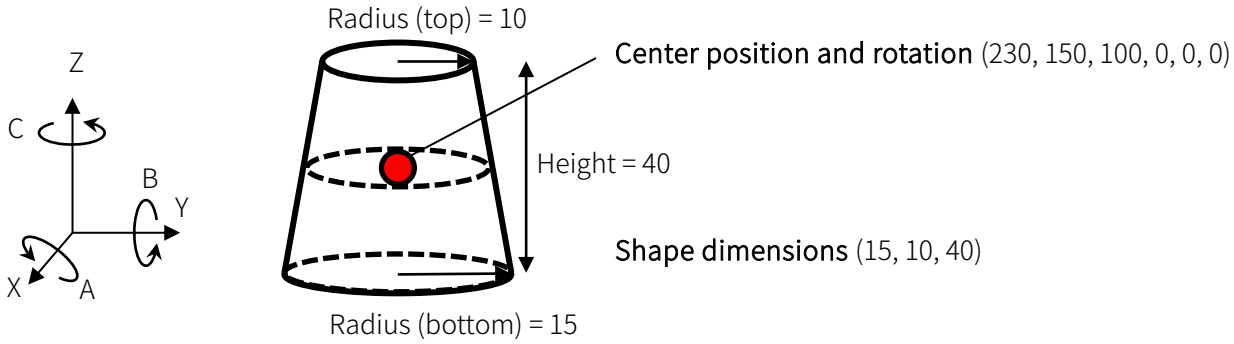


Figure 3.6 Example of setting a cylindrical area

Type 3: Sphere

Set the area using a sphere created by specifying the center position and shape dimension.
Set the coordinates of the center position and rotation with AREA[num]. DEFINITION1 and the shape dimensions with AREA[num]. DEFINITION2.

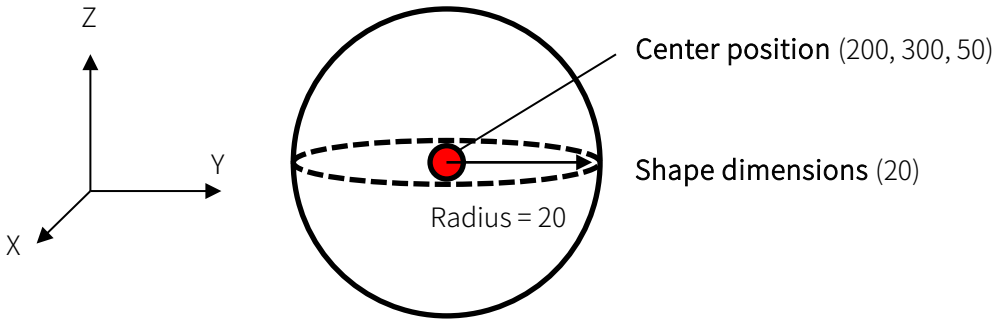


Figure 3.7 Example of setting a spherical area

Output signal

Specify DOUT No. to be output when the robot tip intrudes into the area.
Set a positive value when the signal is turned ON in the area.
Set a negative value when the signal is turned OFF in the area.

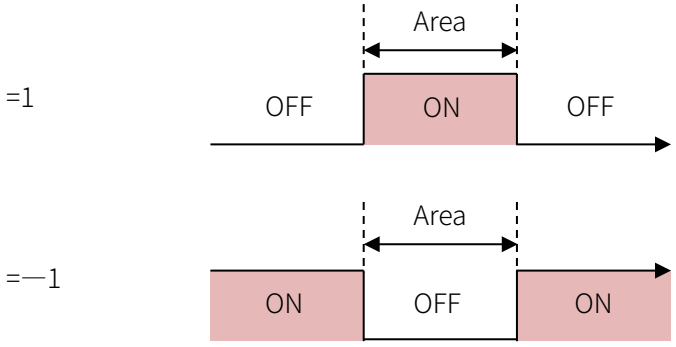


Figure 3.8 Example of setting output signal

[A000-num] AREA TABLE

TABLE

Up to 64 settings can be saved.

Parameter key	AREA[num[1)]].USE-FLG	
Parameter data	=(Disabled/Enabled)	
Data type	Integer type	
Number of data items	1	
Data range	0 or 1	
Description	=0	Disable the setting in AREA[num].
	=1	Enable the setting in AREA[num].
Default	=0	

Parameter key	AREA[num[1)]]. TYPE	
Parameter data	=(Area type)	
Data type	Integer type	
Number of data items	1	
Data range	0 to 3	
Description	= 0	Set a 2-point cuboid area.
	= 1	Set a cuboid area.
	= 2	Set a cylindrical area.
	= 3	Set a spherical area.
Default	= 0	
System version	X8KCS-02C or later	

Parameter key	AREA[num[1)]]. DEFINITION1	
Parameter data	Type 0:	=(X[mm]), (Y[mm]), (Z[mm]), (-), (-), (-), (-), (-)
	Type 1-3:	=(X[mm]), (Y[mm]), (Z[mm]), (A[deg]), (B[deg]), (C[deg]), (-), (-)
Data type	Real number type	
Number of data items	8	
Data range	-9999.999 to 9999.999	
Description	TYPE: 0	Set point 1 that specifies the area.
	TYPE: 1-3	Set the center position and rotation that specify the area.
Default	= 9999.0, 9999.0, 9999.0, 0.000000, 0.000000, 0.000000, 0.000000, 0.000000	

Parameter key	AREA[num[1)]]. DEFINITION2	
Parameter data	Type 0:	=(X[mm]), (Y[mm]), (Z[mm]), (-), (-), (-), (-), (-)
	Type 1:	=(Depth [mm]), (Width [mm]), (Height [mm]), (-), (-), (-), (-), (-)
	Type 2:	=(Radius (bottom) [mm]), (Radius (top) [mm]), (Height [mm]), (-), (-), (-), (-), (-)
	Type 3:	=(Radius [mm]) , (-), (-), (-), (-), (-), (-), (-)
Data type	Real number type	
Number of data items	8	
Data range	-9999.999 to 9999.999	
Description	Type: 0	Set point 2 that specifies the area.
	Type: 1-3	Set the shape dimensions that specify the area.
Default	=-9999.0, -9999.0, -9999.0, 0.000000, 0.000000, 0.000000, 0.000000, 0.000000	

Parameter key	AREA[num[1)]].DOUT-NO
Parameter data	=(External output signal)
Data type	Integer type
Number of data items	1
Data range	External output signal (output signal that can be specified in DOUT command)
Description	Set a positive value when output is ON in the area. Set a negative value when output is OFF in the area.
Default	=num+1

1) num takes an integer from 0 to 63.

Setting example

```
'[A000] AREA OUTPUT SETTING PARAMETER
'{Area Use Flag (0:Disable 1:Enable)}
'{Area Type (0:2-Point, 1:Box, 2:Cylinder, 3:Sphere)}
'{Area Definition1}
'{Area Definition2}
'{Dout Number}

'{Table No.1}
AREA[0]. USE-FLG = 1
AREA[0]. AREA-TYPE = 0
AREA[0]. DEFINITION1= 500, 200.0, 200, 0.000000, 0.000000, 0.000000, 0.000000, 0.000000
AREA[0]. DEFINITION2= 0.000000, -200.0, 0.000000, 0.000000, 0.000000, 0.000000, 0.000000, 0.000000
AREA[0]. DOUT-NO = 1
    •
    •
    •
'{Table No.64}
AREA[63]. USE-FLG = 0
AREA[63]. AREA-TYPE = 0
AREA[63]. DEFINITION1= 9999.9, 9999.9, 9999.9, 0.000000, 0.000000, 0.000000, 0.000000, 0.000000
AREA[63]. DEFINITION2= -9999.9, -9999.9, -9999.9, 0.000000, 0.000000, 0.000000, 0.000000, 0.000000
AREA[63]. DOUT-NO = 64
```

MEMO

- *DOUT assigned by this parameter cannot be operated using SCOL2 language and teach pendant.*
- *This parameter is enabled regardless of whether the program is executed.*
- *When the same output signal is specified in multiple tables (including mixing of output ON/OFF), the output with the larger area number is given priority in overlapping areas. If the output signal is not in any of the areas, the output with the smallest area number will be inverted.*

[A001] RESTRICTED AREA

Set a restricted area

Set an area where the robot can move.

The robot is stopped when the TCP, tool flange, or elbow joint (2nd axis) center point comes out from the area specified in [A000]. Returning from the stopped state can be performed by JOG.

Parameter key	RESTRICTED-AREA-NO	
Parameter data	=(Number of the area to be specified)	
Data type	Integer type	
Number of data items	1	
Data range	0 to 64	
Description	= 0	Disable the setting.
	= 1 - 64	Enable the area with the specified area number.
Default	= 0	
System version	X8KCS-02C or later	

Parameter key	RESTRICTED-USE-TCP	
Parameter data	=(Disabled/Enabled)	
Data type	Integer type	
Number of data items	1	
Data range	0 or 1	
Description	= 0	Disable TCP for check of the movable range.
	= 1	Enable TCP for check of the movable range.
Default	= 1	
System version	X8KCS-02C or later	

Parameter key	RESTRICTED-USE-MIF	
Parameter data	=(Disabled/Enabled)	
Data type	Integer type	
Number of data items	1	
Data range	0 or 1	
Description	= 0	Disable the tool flange for check of the movable range.
	= 1	Enable the tool flange for check of the movable range.
Default	= 1	
System version	X8KCS-02C or later	

Parameter key	RESTRICTED-USE-ELBOW	
Parameter data	=(Disabled/Enabled)	
Data type	Integer type	
Number of data items	1	
Data range	0 or 1	
Description	= 0	Disable the elbow joint (2nd axis) for check of the movable range.
	= 1	Enable the elbow joint (2nd axis) for check of the movable range.
Default	= 1	
System version	X8KCS-02C or later	

Setting example

```
'[A001] RESTRICTED AREA
'{Area Number (0:Disable)}
RESTRICTED-AREA-NO = 3
'{Check TCP Flag (0:Disable 1:Enable)}
RESTRICTED-USE-TCP = 1
'{Check M-I/F Flag (0:Disable 1:Enable)}
RESTRICTED-USE-MIF = 0
'{Check Elbow Flag (0:Disable 1:Enable)}
RESTRICTED-USE-ELBOW = 0
```

MEMO

- *It is necessary to enable the area to be specified. When output of an external signal is not performed, set the output signal (AREA[num].DOUT-NO) to 0.*

[A002] NO ENTRY AREA

Setting a no-entry area

Set an area where the robot cannot enter. Up to four areas can be set.
The robot is stopped when the TCP, tool flange, or elbow joint (2nd axis) center point enters the area specified in [A000]. Returning from the stopped state can be performed by JOG.

Parameter key	NOENTRY-AREA-NO	
Parameter data	=(Number of the 1st area to be specified), (2nd area to be specified), (3rd area to be specified), (4th area to be specified)	
Data type	Integer type	
Number of data items	4	
Data range	0 to 64	
Description	= 0	Disable the setting.
	= 1 - 64	Enable the area with the specified area number.
Default	= 0, 0, 0, 0	
System version	X8KCS-02C or later	

Parameter key	NOENTRY-USE-TCP	
Parameter data	=(Disabled/Enabled), (Disabled/Enabled), (Disabled/Enabled), (Disabled/Enabled)	
Data type	Integer type	
Number of data items	4	
Data range	0 or 1	
Description	= 0	Disable TCP for check of the movable range.
	= 1	Enable TCP for check of the movable range.
Default	= 1, 1, 1, 1	
System version	X8KCS-02C or later	

Parameter key	NOENTRY-USE-MIF	
Parameter data	=(Disabled/Enabled), (Disabled/Enabled), (Disabled/Enabled), (Disabled/Enabled)	
Data type	Integer type	
Number of data items	4	
Data range	0 or 1	
Description	= 0	Disable the tool flange for check of the movable range.
	= 1	Enable the tool flange for check of the movable range.
Default	= 1, 1, 1, 1	
System version	X8KCS-02C or later	

Parameter key	NOENTRY-USE-ELBOW	
Parameter data	=(Disabled/Enabled), (Disabled/Enabled), (Disabled/Enabled), (Disabled/Enabled)	
Data type	Integer type	
Number of data items	4	
Data range	0 or 1	
Description	= 0	Disable the elbow joint (2nd axis) for check of the movable range.
	= 1	Enable the elbow joint (2nd axis) for check of the movable range.
Default	= 1, 1, 1, 1	
System version	X8KCS-02C or later	

Setting example

```
'[A002] NO ENTRY AREA
'{Area Number (0:Disable)}
NOENTRY-AREA-NO = 2, 5, 8, 0
'{Check TCP Flag (0:Disable 1:Enable)}
NOENTRY-USE-TCP = 1, 0, 0, 0
'{Check M-I/F Flag (0:Disable 1:Enable)}
NOENTRY-USE-MIF = 0, 0, 0, 0
'{Check Elbow Flag (0:Disable 1:Enable)}
NOENTRY-USE-ELBOW = 0, 0, 1, 0
```

MEMO

- *It is necessary to enable the area to be specified. When output of an external signal is not performed, set the output signal (AREA[num].DOUT-NO) to 0.*

CONVEYOR.PAR

This chapter explains CONVEYOR.PAR.

Make a setting to operate the robot and conveyor in cooperation using CONVEYOR.PAR.

To use the vision and conveyor together, refer to STA1183_VISION SENSOR + CONVER SYNCHRONIZATION FUNCTION MANUAL.

[C000] CONVEYOR PARAMETER

The conveyor parameter (CONVEYOR.PAR:[C000] Conveyor Parameter) is a parameter exclusively for conveyor synchronization.

C000 has the following setting items.

No.	Data name	Description	Initial value (Ch1 / Ch2)
C000-1	Encoder rotating direction	Encoder rotating direction	0 0
C000-2	Encoder sampling period [msec]	Encoder sampling period	3 3
C000-3	Conveyor stop judgment time [msec]	Conveyor stop judgment time	0 0
C000-4	Type of conveyor synchronization	Type of conveyor synchronization (0: Straight line 1: Arc (forward rotation) 2: Arc (reverse rotation) 3: Vision + conveyor)	0 0
C000-5	Dummy data	Dummy data	0 0
C000-6	Dummy data	Dummy data	0 0
C000-7	Dummy data	Dummy data	0 0
C000-8	Travel value per pulse of encoder[um/pulse]	Travel value per pulse of encoder	0.0 0.0
C000-9	Rotation center position X of arc conveyor [mm]	Rotation center position X of arc conveyor	0.0 0.0
C000-10	Rotation center position Y of arc conveyor [mm]	Rotation center position Y of arc conveyor	0.0 0.0
C000-11	Limitation width of synchronization(inner)	Conveyor synchronization limitation area (inner circumference) (mm)	0.0 0.0
C000-12	Limitation width of synchronization(outer)	Conveyor synchronization limitation area (outer circumference) (mm)	0.0 0.0
C000-13	Dummy data	Dummy data	0.0 0.0
C000-14	Dummy data	Dummy data	0.0 0.0
C000-15	Dummy data	Dummy data	0.0 0.0

[C000-1]Encoder rotating direction

Encoder rotating direction

The encoder rotating direction is determined depending on the specifications of the encoder used and its rotating direction when the encoder is mounted on the conveyor.

Data type	Integer type	
Number of data items	2	
Data range	0 or 1	
Description	=0	Encoder rotating direction is positive
	=1	Encoder rotating direction is negative
Default	=0 , 0	

Example of setting vision + conveyor function

```
'{Encoder rotating direction}
CONV-DIRECTION = 0, 0
```

MEMO

For details about the direction of rotation, refer to the attached "STA1183 Vision + Conveyor Synchronization Function Manual".

[C000-2]Encoder sampling period [msec]

Encoder sampling period

Set the period of sampling conveyor encoder counter value.

Data type	Integer type
Number of data items	2
Data range	3 to 30000
Unit of data	ms
Default	=3 , 3

Example of setting vision + conveyor function

```
'{Encoder sampling period [ms]}
CONV-SMP-PERIOD = 3, 3
```

MEMO

By default, the conveyor synchronization function calculates the synchronous travel value with a minimum cycle of 3 ms, but it is possible to specify a larger value. Keep the following in mind when making adjustments.

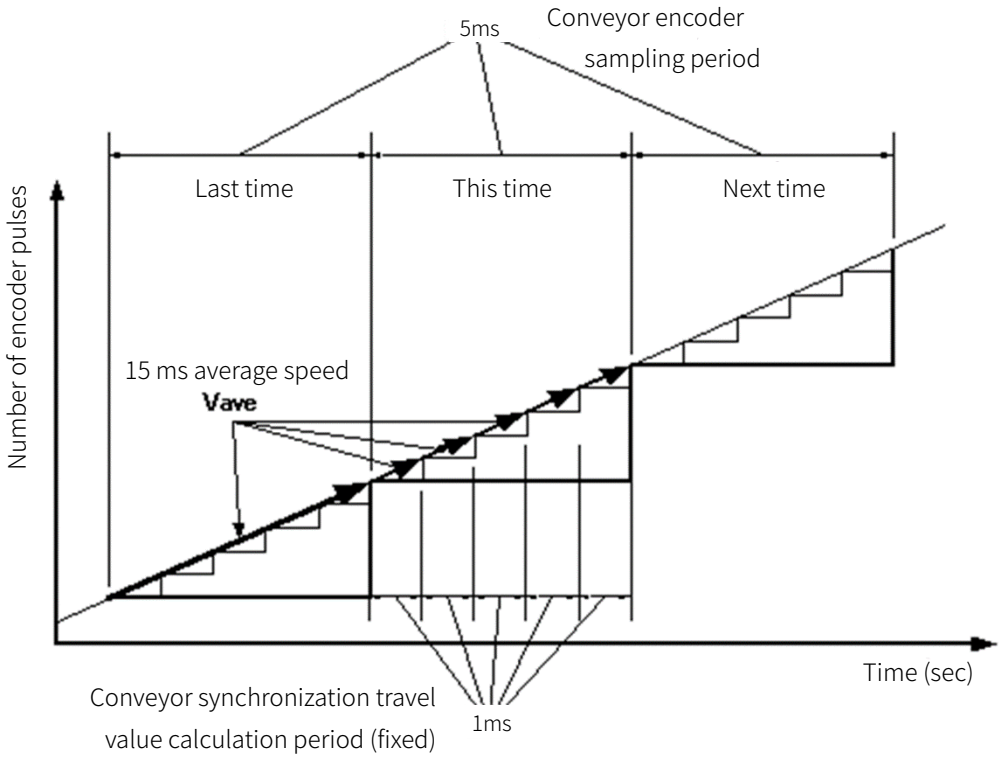
■ *When this parameter is set to a smaller value (3 ms at a minimum)*

- *The tracking performance with respect to the conveyor speed is improved.*
- *If the parameter is specified to a too small value, pulsation of the conveyor is followed sensitively, which may cause the robot to vibrate.*

■ *When this parameter is set to a larger value*

- *The susceptibility to pulsation can be reduced.*
- *Calculation errors of the conveyor rotating angle can be absorbed.*
- *The tracking performance with respect to the conveyor speed is deteriorated.*
- *When no pulsation exists, specify the parameter to the default value (3 ms).*

(Example) When the conveyor sampling period is set to 5 ms, the synchronous traveling value is calculated based on the average speed of the previous conveyor sampling period "Vave".



[C000-3]Conveyor stop judgment time [msec]

Time to determine that the conveyor is stopped

Specify the time to determine that the conveyor is stopped. If the encoder counter does not change during the time specified here, the conveyor is judged to be stopped, and the error "001-048: Conveyor Stop" occurs. Note that this error is a level 1 error, so program operation continues without stopping.

Data type	Integer type
Number of data items	2
Data range	0 to 30000
Unit of data	ms
Default	=0 , 0

Example of setting vision + conveyor function

```
'{Conveyor stop judgment time [ms]}
CONV-TIMEOUT = 0, 0
```

MEMO

- *Normally, the time to judge a stoppage is specified by the second argument in the SCOL2 language "SYNC" command, but if this is omitted, a conveyor stop is judged by this set value.*
- *If the second argument of the SYNC command is omitted and this parameter is set to "0", stop judgment is not performed.*

[C000-4]Type of conveyor synchronization

Type of conveyor synchronization

Specify the type of conveyor synchronization function.
The type can be selected from two types: Straight conveyor synchronization function and vision + conveyor synchronization function.

When selecting the vision + conveyor synchronization function, set this parameter to “3”.

Data type	Integer type	
Number of data items	2	
Data range	0 to 3	
Description	=0	Straight conveyor (can be set for CH1 and CH2)
	=1	(Reserved)
	=2	(Reserved)
	=3	Vision + conveyor synchronization function (can be set for CH1 and CH2)
Default	=0 , 0	

Example of setting vision + conveyor function

```
'{Type of conveyor synchronization}
CONV-FORM = 3, 0
```

MEMO

- *If this parameter is not set, the error "001-0147: COM Buffer Over" occurs.*

[C000-5]Dummy data

[C000-6]Dummy data

[C000-7]Dummy data

Spare area

Spare areas for the system

Data type	Integer type
Number of data items	2
Default	=0 , 0

Default setting

```
'{Dummy data}
CONV-ERR-COMP = 0, 0
'{Dummy data}
CONV-ERR-COMP-VELGAIN = 0, 0
'{Dummy data}
CONV-ERR-COMP-ACCGAIN = 0, 0
```

MEMO

- *Do not change this setting.*

[C000-8]Travel value per pulse of encoder[um/pulse]

Travel value per pulse of encoder

Obtain the conveyor travel value[1] (μm) per pulse of encoder.

Data type	Real number type
Number of data items	2
Unit of data	μm
Default	=0 , 0

Example of setting vision + conveyor function

```
'{Travel value per pulse of encoder [um/pulse] [mdeg/pulse]}
CONV-ENC-GEAR = 12.000000, 0.000000
```

1) Derivation of travel value per pulse of encoder

Travel value per pulse of encoder α (μm) is obtained by the following equation where the conveyor travel is ΔL (mm) and the total number of encoder pulses is ΔP (pulses).

$$\alpha = (\Delta L / \Delta P) * 1000 \quad ①$$

For details about how to measure the total number of encoder pulses ⊿P , refer to “STA1183_Vision + Conveyor Synchronization Function Manual.”

[C000-9]Rotation center position X of arc conveyor [mm]

[C000-10]Rotation center position Y of arc conveyor [mm]

X and Y coordinates of arc conveyor rotation center position

Set this parameter when an arc conveyor[1)] is used.

1) TS5000 does not support this function.

Default setting

```
'{Rotation center position X of arc conveyor [mm]}
CONV-CIRC-XCENT = 0.000000, 0.000000
'{Rotation center position Y of arc conveyor [mm]}
CONV-CIRC-YCENT = 0.000000, 0.000000
```

[C000-11]Limitation width of synchronization(inner)

[C000-12]Limitation width of synchronization(outer)

Limitation area of conveyor synchronization

This parameter is specified by the distance from the inner limitation circle and outer limitation circle of the robot movement.

Data type	Real number type
Number of data items	2
Unit of data	mm
Default	=0 , 0

Example of setting vision + conveyor function

```
'{Limitation width of synchronization(inner)}
CONV-SLIM-WIDTH[0] = 50.000000, 0.000000
'{Limitation width of synchronization(outer)}
CONV-SLIM-WIDTH[1] = 600.000000, 0.000000
```

MEMO

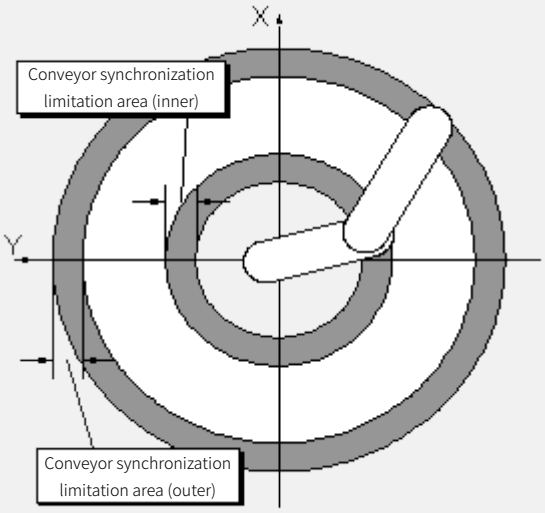
・If the robot enters this conveyor synchronization limitation area during conveyor synchronization, the "004-051: Conveyor sync limit area" error will occur, causing the robot to slow down and stop. At the same time, the conveyor synchronization will stop.

・Limitation width of synchronization(inner)

→Distance from the inner limitation circle of the robot movement

・Limitation width of synchronization(outer)

→Distance from the outer limitation circle of the robot movement



[C000-13]Dummy data

[C000-14]Dummy data

[C000-15]Dummy data

Spare area

Spare areas for the system

Data type	Real number type
Number of data items	2
Default	=0.000000, 0.000000

Default setting

```
'{Dummy data}
CONV-DUMY-F[0] = 0.000000, 0.000000
'{Dummy data}
CONV-DUMY-F[1] = 0.000000, 0.000000
'{Dummy data}
CONV-DUMY-F[2] = 0.000000, 0.000000
```

MEMO

- *Do not change this setting.*

[C001] VISION + CONVEYOR TRACKING PARAMETER (CH1,CH2)

This parameter is dedicated to the vision + conveyor synchronization function. This parameter is enabled only when "3: Vision + conveyor synchronization function" is set in "Type of conveyor synchronization" of the conveyor parameter [C000-4] mentioned above.

To use the vision and conveyor together, refer to STA1183_VISION + CONVEYOR SYNCHRONIZATION MANUAL.

No.	Data name	Description	Initial value (Ch1 / Ch2)	
C001-1	Delimiter Character	Communication data delimiter (0: Comma)	0	0
C001-2	Work sort function enable SW	Enable/disable the data sort function	0	0
C001-3	Virtual vision conveyor tracking enable SW[1)]		0	0
C001-4	Data reception timeout [ms]	Capture data reception timeout [ms]	0.0	0.0
C001-5	Calibration offset [mm]	Calibration offset [mm]	0.0	0.0
C001-6	Detection area start point [mm]	Workpiece detection area start point [mm]	0.0	0.0
C001-7	Detection area end point [mm]	Workpiece detection area end point [mm]	0.0	0.0
C001-8	Conveyor Sync limit point [mm]	Conveyor synchronization limit point [mm]	0.0	0.0
C001-9	Overlap allowance [mm]	Workpiece overlap allowance [mm]	0.0	0.0
C001-10	Capture width [mm]	Capture width [mm]	0.0	0.0
C001-11	Calibration reference point (X)	Calibration reference position X coordinate [mm]	0.0	0.0
C001-12	Calibration reference point (Y)	Calibration reference position Y coordinate [mm]	0.0	0.0
C001-13	Minimum Trigger Interval [msec]	Trigger detection minimum interval [ms]	0.0	0.0

1) TS5000 does not support this function.

[C001-1]Delimiter Character

Communication data delimiter character

Specify the delimiter character of communication data.

Data type	Real number type	
Number of data items	2	
Description	=0	, (Comma)
	=1	(Space)
Default	=0, 0	

Default setting

'{Delimiter Character (0:' , ' 1:' ')}
VCV-DLMT-CHAR = 0, 0

MEMO

- *The delimiter character of communication data is always "," (comma), so this parameter is not used.*

[C001-2]Work sort function enable SW

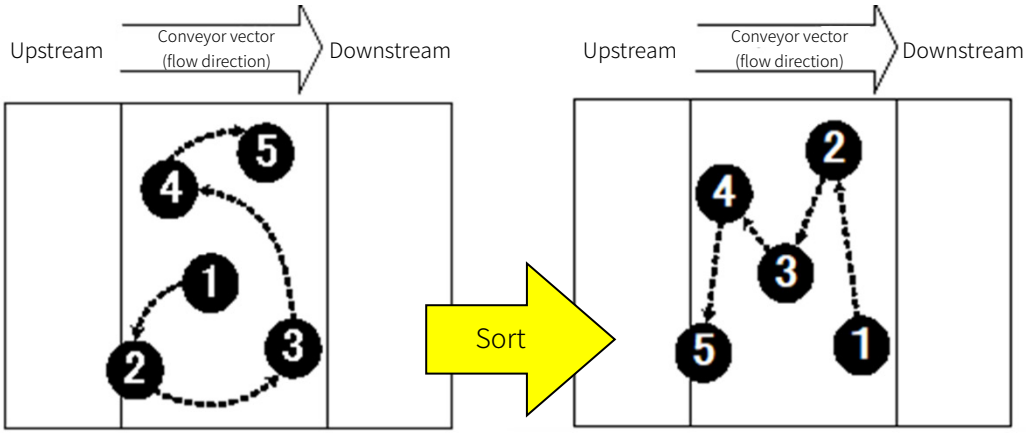
Data sort function

This function automatically sorts the order of received workpiece data items from downstream to upstream of the conveyor. When this function is enabled, workpieces can be processed in order from the downstream, so optimum picking operation with less waste can be performed.

Data type	Integer type	
Number of data items	2	
Description	=0	Disable
	=1	Enable
Default	=0, 0	

Setting

```
'{Work sort function enable SW (0:Disable, 1:Enable)}
VCV-WSORT-SW = 1, 0
```



[C001-3]Virtual vision conveyor tracking enable SW[1)]

Virtual vision coveyor synchronization function[1)]

1) TS5000 does not support this function.

Data type	Integer type	
Number of data items	2	
Description	=0	Disable
	=1	Enable
Default	=0, 0	

Default setting

```
'{Virtual vision conveyor tracking enable SW (0:Disable, 1:Enable)}
VCV-VRVCVENABLE = 0, 0
```

MEMO

- *Do not change this setting.*

[C001-4]Data reception timeout [ms]

Capture data reception timeout

This function issues the "1-180: VCV Comm Process Timeout" error when data reception is not completed within the specified time.

Data type	Real number type
Number of data items	2
Unit of data	ms
Default	= 0.000000, 0.000000

Default setting

```
'{Data reception timeout [ms]}
VCV-RCV-TIMEOUT = 0.000000, 0.000000
```

MEMO

- *When the conveyor parameter [C001] {Data reception timeout [ms]} of this function is set to "0.000000", this function is disabled.*

[C001-5]Calibration offset [mm]

Calibration offset

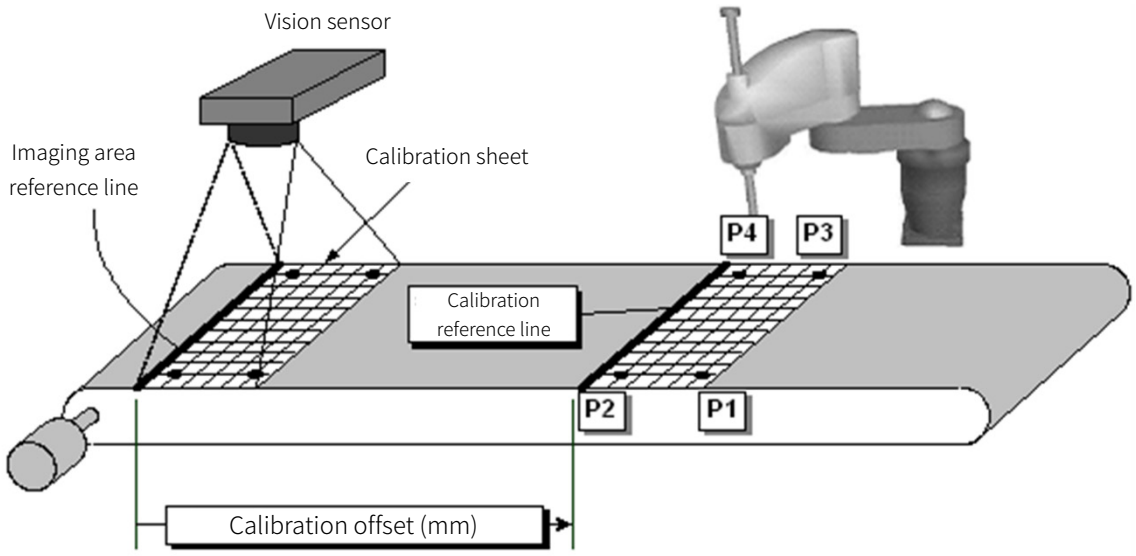
This parameter specifies the distance from the imaging area reference line to the calibration[1)] reference line in millimeters.

Data type	Real number type
Number of data items	2
Unit of data	mm
Default	= 0.000000, 0.000000

Setting example

```
'{Calibration offset [mm]}
VCV-CALIB-OFST = 425.000000, 0.000000
```

1) Calibration converts the coordinate system of the vision sensor to the coordinate system of the robot. When calibrated by the vision sensor, all the coordinate values transmitted by the vision sensor are converted to values based on the world coordinate system of the robot.



[C001-6]Detection area start point [mm]

[C001-7]Detection area end point [mm]

[C001-8]Conveyor Sync limit point [mm]

Workpiece detection area start point

Specify the distance from the capture origin to the workpiece detection area[1)] start point in millimeters.

Workpiece detection area end point

Specify the distance from the capture origin to the workpiece detection area[1)] end point in millimeters.

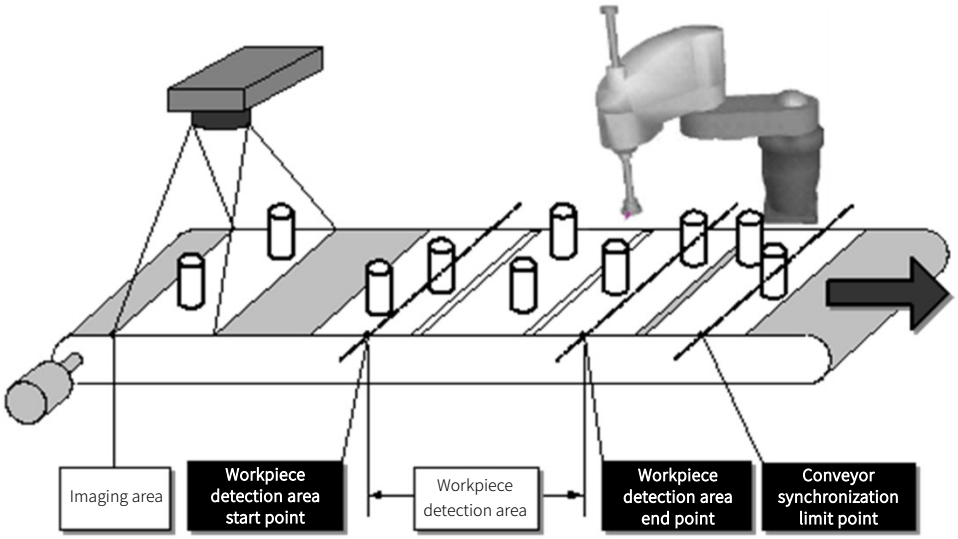
Conveyor synchronization limit point

Specify the distance from the capture origin to the conveyor synchronization limit point in millimeters.

Data type	Real number type
Number of data items	2
Unit of data	mm
Default	= 0.000000, 0.000000

Setting example

```
'{Detection area start position [mm]}
VCV-DETECT-STR = 425.000000, 0.000000
'{Detection area end position [mm]}
VCV-DETECT-END = 650.000000, 0.000000
'{Conveyor Sync limit position [mm]}
VCV-SYNC-LIMIT = 850.000000, 0.000000
```



■ Workpiece detection area

When a workpiece on the conveyor enters the workpiece detection area, the robot starts following the conveyor.

At the same time, the robot starts to move to the position of the most downstream workpiece among the workpieces in the workpiece detection area.

When no workpiece is found in the workpiece detection area, synchronization does not start and the system waits until the next workpiece enters the detection area.

■ Conveyor synchronization limit point

After the robot synchronization starts, the conveyor synchronization ends when the robot tip position is beyond the conveyor synchronization limit position.

NOTICE

- If the speed of the robot speed is slow, the robot makes motion beyond the conveyor synchronization limit point and collides with a peripheral device. Be careful when the speed of the robot is slow.

[C001-9]Overlap allowance [mm]

Overlap allowance

When the distance ⊿w between the workpiece detected last time and that detected currently is less than the specified value (L), the workpiece detected currently is ignored.

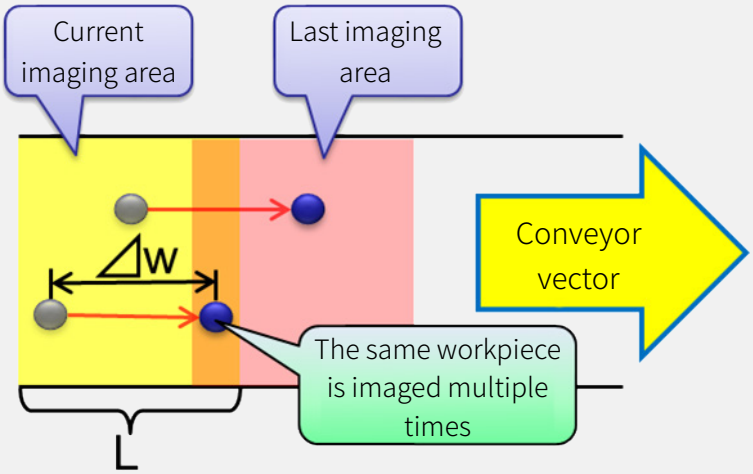
Data type	Real number type
Number of data items	2
Unit of data	mm
Default	= 0.000000, 0.000000

Setting example

```
'{Overlap allowance [mm]}
VCV-OVERLAP-ALW = 30.000000, 0.000000
```

MEMO

- *Use this function when the same workpiece has been imaged multiple times as shown in the figure below.*



[C001-10]Capture width [mm]

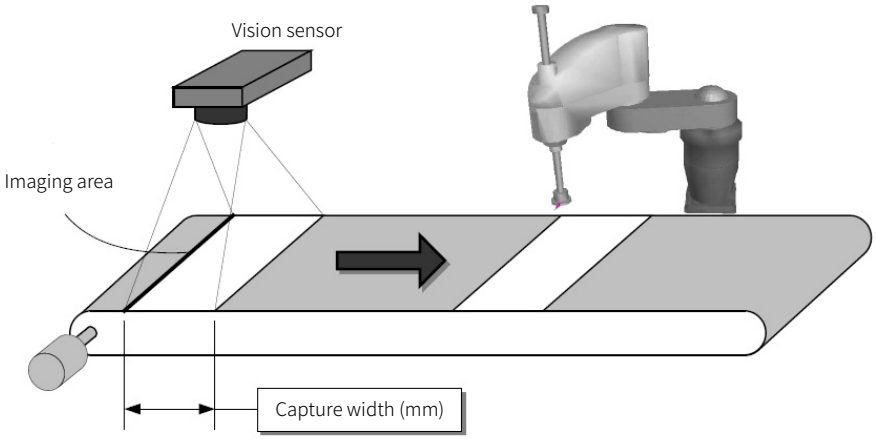
Capture width

Specify the width (mm) of images to be captured by the vision sensor.

Data type	Real number type
Number of data items	2
Unit of data	mm
Default	= 0.000000, 0.000000

Setting example

```
'{Capture width [mm]}
VCV-CAP-WIDTH = 170.000000, 0.000000
```



[C001-11]Calibration reference point (X)

[C001-12]Calibration reference point (Y)

Calibration reference position X coordinate

Calibration reference position Y coordinate

The calibration reference position (X, Y) is an arbitrary world coordinate system position on the calibration reference line. By registering this reference position, the robot controller recognizes the relationship between the vision sensor imaging area reference line and the calibration position.

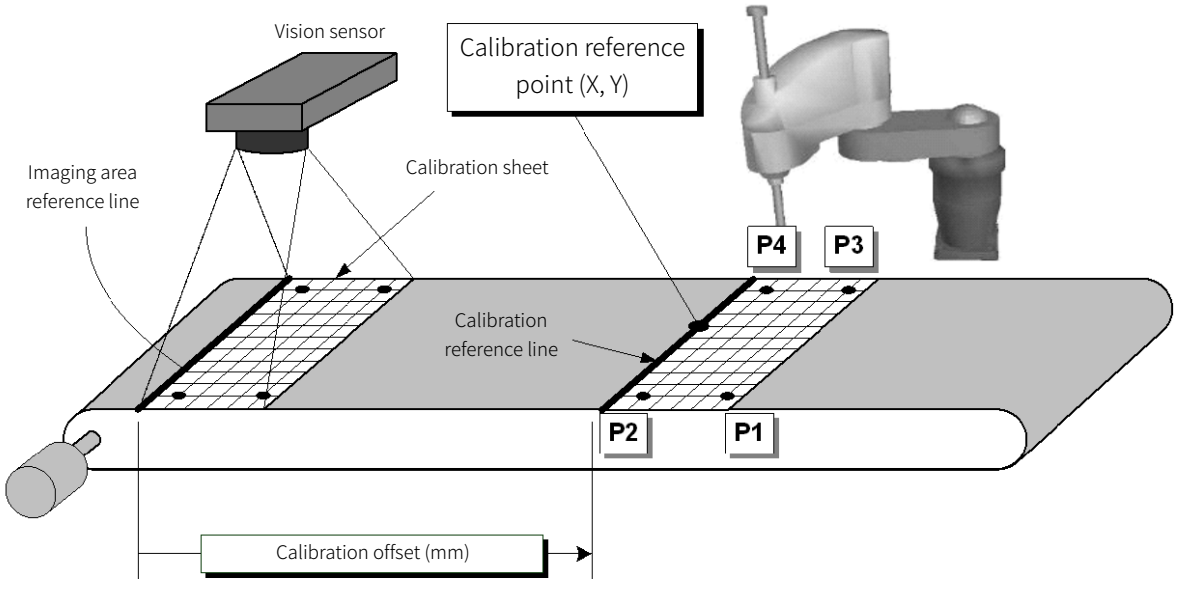
Data type	Real number type
Number of data items	2
Unit of data	mm
Default	= 0.000000, 0.000000

Setting example

```
'{Calibration reference point (X)}
VCV-CAP-ORG-X = 450.000000, 0.000000
'{Calibration reference point (Y)}
VCV-CAP-ORG-Y = -290.000000, 0.000000
```

MEMO

- *Specify the calibration reference position (X, Y) with the position X, Y (mm) in the world coordinate system of the robot. (For details about the setting method, refer to "STA1183_Vision + Conveyor Synchronization Function Manual.")*



[C001-13]Minimum Trigger Interval [msec]

Minimum trigger interval

Data type	Real number type
Number of data items	2
Unit of data	ms
Default	= 0.000000, 0.000000

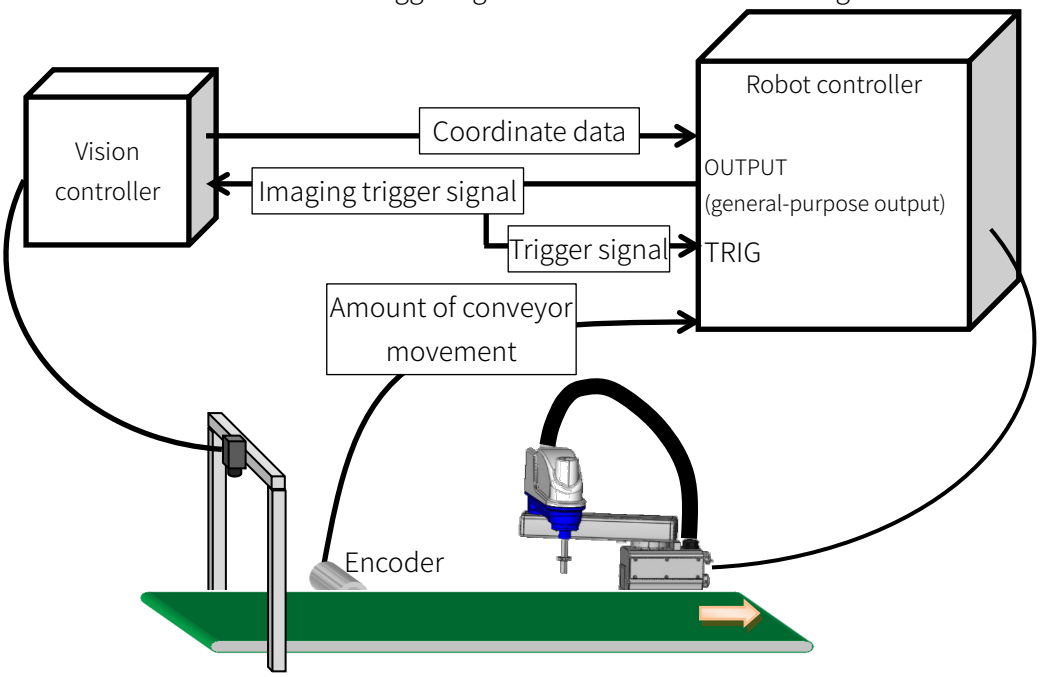
Setting example

```
'{Minimum Trigger Interval [ms]}
VCV-MINTRIGINTERVAL = 0.000000, 0.000000
```

[C002] CONVEYOR TRACKING DOUT FUNCTION

This parameter is to set the conveyor synchronous pulse output function. This parameter becomes enabled regardless of which conveyor synchronization type is set in "Type of conveyor synchronization" of the conveyor parameter [C000] mentioned above.

This function is used when the trigger signal is wired as shown in the figure below.



No.	Data name	Description	Initial value (Ch1 / Ch2)
C002-1	Output signal number	Output signal number	0 0
C002-2	Width of signal ON [mm]	Width of signal ON	0.0 0.0
C002-3	Width of one cycle [mm]	Width of one cycle	0.0 0.0

[C002-1]Output signal number

Output signal number

Specify the signal to be pulse-output as an imaging trigger signal. Specify any general-purpose output signal number.

Data type	Integer type
Number of data items	2
Data range	0 to 25
Default	=0, 0

Setting example

```
'{Output signal number}
CONV-CYCLE-IONO = 1, 0
```

MEMO

- *To use DOUT 9 to 25, use an expansion I/O board.*
- *Setting this parameter to "0" disables the function.*

[C002-2]Width of signal ON [mm]

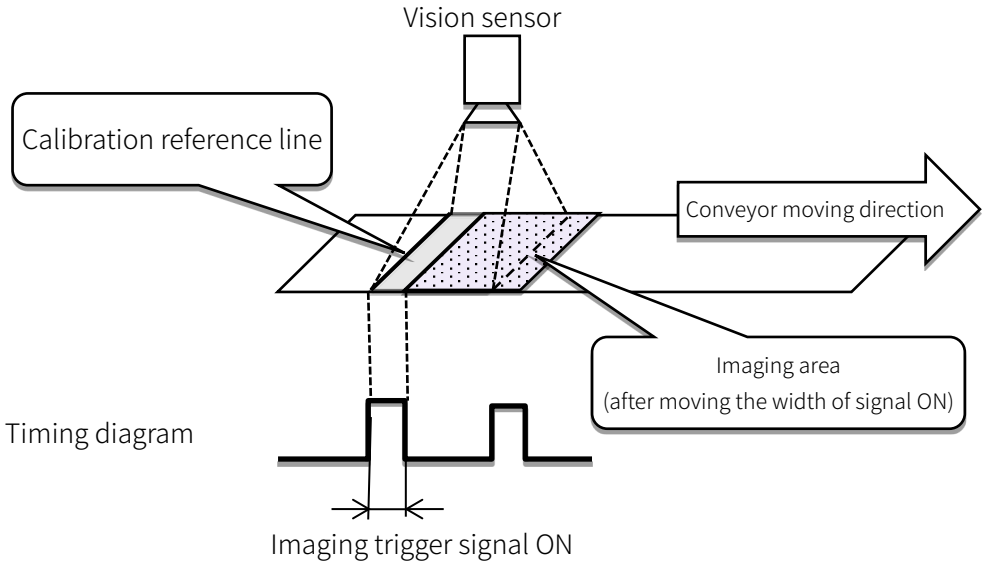
Width of signal ON

Specify the amount of conveyor movement in which the imaging trigger signal is kept ON.

Data type	Real number type
Number of data items	2
Default	=0.000000, 0.000000

Setting example

```
'{Width of signal on [mm]}
CONV-CYCLE-ON-WIDTH = 50.000000, 0.000000
```



[C002-3]Width of one cycle [mm]

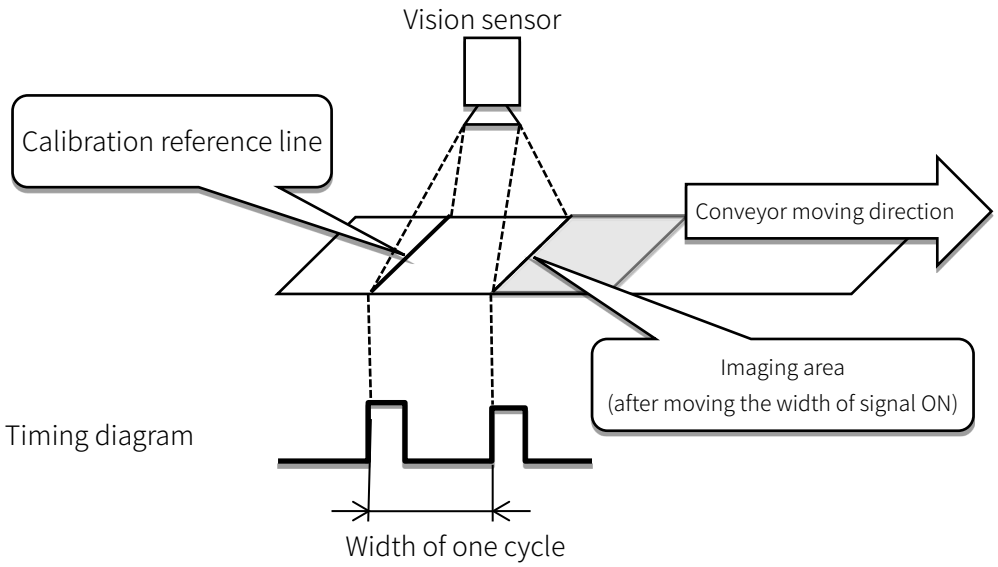
Width of one cycle

Specify the amount of conveyor movement in one cycle of signal output.

Data type	Real number type
Number of data items	2
Default	=0.000000, 0.000000

Setting example

```
'{Width of one cycle [mm]}
CONV-CYCLE-WIDTH = 125.000000, 0.000000
```



[C003] CONVEYOR FUNCTION SWITCH

Do not change this parameter. TS5000 does not support this parameter.

[C004] CONVEYOR FUNCTION SWITCH2

Do not change this parameter. TS5000 does not support this parameter.

[C005] DATA SET GROUP FOR VIRTUAL VISION CONVEYOR TRACKING

Do not change this parameter. TS5000 does not support this parameter.